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Inventor(s)	Lu; Boxun et al.

Compound for treating neurodegenerative disorders

Abstract

The present invention relates to use of a compound with a substituted bicyclic structure, a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating polyglutamine (polyQ) related disorders.

Inventors:	Lu; Boxun (Shanghai, CN), Fei; Yiyang (Shanghai, CN), Ding; Yu (Shanghai, CN), Dang; Yongjun (Shanghai, CN)
Applicant:	FUDAN UNIVERSITY (Shanghai, CN)
Family ID:	72427189
Assignee:	Fudan University (Shanghai, CN)
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Primary Examiner: Coleman; Brenda L

Attorney, Agent or Firm: IceMiller LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. national stage filing, under 35 U.S.C. § 371(c), of International Application No. PCT/CN2020/078779, filed on Mar. 11, 2020, which claims priority to Chinese Patent Application No. 201910180674.7, filed on Mar. 11, 2019 and Chinese Patent Application No. 201911000198.2, filed Oct. 21, 2019, the entire contents of each of which are incorporated herein by reference.

FIELD OF THE INVENTION

(2) The present invention relates to the biopharmaceutical field, and in particular to use of a compound with a substituted bicyclic structure, a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyglutamine (polyQ) related disorder.

BACKGROUND OF THE INVENTION

(3) Neurodegenerative disorder refers to diseases caused by nervous system dysfunction resulted from abnormal death of central neurons. So far, there is no ultimate therapy that can slow down its development. Many neurodegenerative disorders are caused by proteins with unknown activity. Currently, methods that may be used to control protein levels, such as biological tools like RNAi, CRISPR. etc. are difficult to deliver, especially to the nervous system.

(4) A feasible treatment strategy is to use low molecular weight compounds (compounds for short) to control the level of proteins that affect diseases. An emerging approach is to enhance the ubiquitination of disease-causing proteins and target them to the proteasome degradation pathway using proteolysis-targeting chimera (PROTAC) technology, but this method relies on certain E3 ligases, which may not be present in diseased cells. Moreover, the protein degradation ability of the proteasome is limited, and the degradation efficiency of certain large disease proteins or aggregates that cause neurodegenerative disorder is low. As an important protein degradation pathway, autophagy is ubiquitous in eukaryotic cells, with strong protein degradation ability but low selectivity. Some studies increase protein degradation by enhancing autophagy, but this method lacks specificity.

(5) PolyQ-related neurodegenerative disorder is a type of neurodegenerative disorder caused by mutant proteins, which can be effectively treated by reducing the level of mutant proteins. Taking Huntington's disease (HD), the most common neurodegenerative disorder, as an example, it is a monogenetic disorder wherein the mutation in the CAG repeat region of the exon 1 of the HTT gene located in the patient's chromosome 4 results in the expansion of the polyglutamine (polyQ) of the synthesized mutant protein (mHTT). mHTT is susceptible to shearing and aggregating, and causes toxicity which eventually leads to dysfunction and death of specific neuron. The current methods for controlling mHTT levels through low molecular weight compounds lack specificity, and may cause side effects. Furthermore, these methods are nonallele-selective and unable to distinguish between mHTT and wild-type HTT protein (wtHTT), which will lead to a decrease in the level of wtHTT which has important biological functions.

(6) Similarly, take spinocerebellar ataxia type 3 (SCA3; also known as Machado-Joseph disease, MJD) as an example, which is the most common autosomal dominant spinocerebellar ataxia and is second only to HD as a common polyQ-related disorder in the world. SCA3 is caused by the abnormally expanded polyQ at the C-terminus of the coded protein ATXN3 resulting from the increase in the number of CAG repeats of the Ataxin-3 gene (ATXN3; also known as the MJD1 gene). Some studies have used siRNA, antisense oligonucleotides and other means to act on ATXN3 to reduce the expression of mutant ATXN3 protein and it is confirmed that the reduction in the level of mutant ATXN3 protein can produce therapeutic effects (Wang, Neuroscience, 371, 2018, 138-154). Some studies have focused on controlling the level of mutant ATXN3 protein through low molecular weight compounds. Studies on enhancing autophagy by compounds such as Menzies et al. Brain 2010, 133:93-104; studies on reducing the level of mutant ATXN3 protein by

regulating other targets such as Costa M D, Brain, 2016,139(11)2891-2908. But these studies have not satisfactorily solved the problem of specificity.

DISCLOSURE OF THE INVENTION

(7) In one aspect, the present invention provides use of a compound of formula (I), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder

(8) ##STR00001## wherein: ring A is a benzene ring; ring B is a saturated or unsaturated 5- or 6-membered heterocycle, the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S; ring C is selected from C.sub.6-10 aryl and 5- to 10-membered heteroaryl, wherein the aryl or heteroaryl is optionally substituted with one or more substituents each independently selected from R.sup.X1; L.sup.1 is a bond, or C.sub.1-6 a hydrocarbon chain; or ring C is absent, and L.sup.1 is absent; R.sup.1 is =Y, wherein Y is O or S, or OR.sup.7; at each occurrence, R.sup.2 is each independently selected from hydrogen, halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, =O, =S, =NR.sup.a1, —OR.sup.a1, —SR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, —C(=O)NR.sup.a1R.sup.b1, C(=O)R.sup.a1, —S(=O).sub.2OR.sup.a1, —S(=O).sub.2R.sup.a1, —S(=O).sub.2NR.sup.a1R.sup.b1, —S(=O)R.sup.a1, —C(=S)OR.sup.a1, —C(=S)NR.sup.a1R.sup.b1, —C(=S)R.sup.a1, —P(=O)(OR.sup.a1)OR.sup.b1, —C(=NR.sup.a1)NR.sup.b1R.sup.c1, —OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, =O, =S, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2, —C(=O)R.sup.a2, —S(=O).sub.2OR.sup.a2, —S(=O).sub.2R.sup.a2, —S(=O).sub.2NR.sup.a2R.sup.b2, —S(=O)R.sup.a2 and —C(=NR.sup.a2)NR.sup.b2R.sup.c2; R.sup.3, R.sup.4, R.sup.5, R.sup.6 are each independently selected from H and R.sup.X2; at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, —OR.sup.7, —SR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —C(=O)R.sup.7, —S(=O).sub.2OR.sup.7, —S(=O).sub.2R.sup.7, —S(=O).sub.2NR.sup.7R.sup.8, —OS(=O).sub.2R.sup.7, —NS(=O).sub.2R.sup.7R.sup.8 and —S(=O)R.sup.7, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C.sub.1-6 alkyl), —S(C.sub.3-6 cyclohydrocarbyl), —S(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, =O, —COOH and C.sub.1-6 alkyl; at each occurrence, R.sup.7, R.sup.8 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl and C.sub.6-10 aryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl or aryl are optionally substituted with one or more substituents selected from halogen,

—NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl); at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3 to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, —OR.sup.Y1, —SR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2, —C(=O)R.sup.Y1, —S(=O).sub.2OR.sup.Y1, —S(=O).sub.2R.sup.Y1, —S(=O).sub.2NR.sup.Y1R.sup.Y2 and —S(=O)R.sup.Y1, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, =O, =S, —OR.sup.Y3, —SR.sup.Y3, —NR.sup.Y3R.sup.Y4, —C(=O)R.sup.Y3, —C(=O)OR.sup.Y3 and —C(=O)NR.sup.Y3R.sup.Y4; at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-8 alkyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3 to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents independently selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, —OH, —SH, —NH.sub.2, =O and —COOH; n is 1 or 2; provided that the compound of formula (I) does not include the following structure:

(9) ##STR00002## wherein R.sup.3 is selected from —O(C.sub.1-6 alkyl) and —O(C.sub.1-4 alkylene-C.sub.6-10 aryl); R.sup.10 is H; and the compound of formula (I) does not include the following structure:

(10) ##STR00003## wherein R.sup.4 is —O(C.sub.1-6 alkyl); R.sup.5 is halogen; R.sup.10 is CF.sub.3; and the compound of formula (I) does not include the following structure:

(11) ##STR00004## wherein R.sup.4 is selected from —O(C.sub.1-6 alkyl) and —O(C.sub.1-4 alkylene-C.sub.6-10 aryl); ring C is C.sub.6-10 aryl, which is optionally substituted with one or more substituents each independently selected from R.sup.X1; and the compound of formula (I) does not include the following structure:

(12) ##STR00005## wherein R.sup.4 is selected from —OH and —O(C.sub.1-6 alkyl); R.sup.5 is halogen; Ring C is 5- to 10-membered heteroaryl, which is optionally substituted with one or more substituents each independently selected from R.sup.X1.

(13) In one embodiment, the neurodegenerative disorder is spinocerebellar ataxia (such as type 1, 2, 3, 6, 7, 12, 17), dentatorubral-pallidoluysian atrophy, Huntington's disease, Huntington's disease-like syndrome-2 or spinal-bulbar muscular atrophy, especially Huntington's disease.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1. Effects of compounds on the level of proteins containing polyglutamine in HEK293T cells.

(2) FIG. 2. Affinity binding curves determined by OI-RD of compounds with full-length HTT of different concentrations. The vertical dashed lines indicate the beginning of the association phase and the dissociation phase during the affinity binding. The dashed curve is the result of the global fitting of the Langmuir reaction model.

(3) FIG. 3. Affinity binding curve determined by MST of compound with full-length HTT.

- (4) FIG. 4. Effects of compounds on HTT levels in cortical neurons of Hd.h.sup.Q140/Q7 mice.
- (5) FIG. 5. Effects of compound 1 on HTT levels in cortical neurons of Hd.h.sup.Q7/Q7 mice.
- (6) FIG. 6. Detection of N-terminal fragments of mHTT with antibodies MW1 and 3B5H10.
- (7) FIG. 7. Cell viability test results of Hd.h.sup.Q140/Q7 mouse cortical neurons after treatment with the compounds.
- (8) FIG. 8. Effects of compounds on mHTT levels in primary fibroblasts of HD patients at the concentration of 100 nM.
- (9) FIG. 9. Effects of compounds on mHTT levels in immortalized fibroblasts of HD patients.
- (10) FIG. 10. Compound 2 reduces mHTT levels in immortalized fibroblasts of HD patients.
- (11) FIG. 11. Effects of compounds on mHTT levels in HD patient iPSC-derived neurons.
- (12) FIG. 12. Effects of compounds on the apoptosis of HD patient iPSC-derived neurons. The scale bar is 50 μ m.
- (13) FIG. 13. Effects of compounds on the apoptosis of HD patient iPSC-derived neurons.
- (14) FIG. 14. Effects of compounds on mHTT levels in Huntington's disease *Drosophila*.
- (15) FIG. 15. Effects of compounds on the survival rate of Huntington's disease *Drosophila*.
- (16) FIG. 16. Effects of compounds on the climbing performance of Huntington's disease *Drosophila*.
- (17) FIG. 17. Effects of intracerebroventricular injection of compounds on the levels of mHTT and wtHTT in the cortices of Huntington's disease mice.
- (18) FIG. 18. Effects of intraperitoneal injection of compound 2 on the levels of mHTT and wtHTT in the cortices of Huntington's disease mice.
- (19) FIG. 19. Effects of intraperitoneal injection of compound 2 on the levels of mHTT and wtHTT in the striata of Huntington's disease mice.
- (20) FIG. 20. Detection of mHTT aggregates in the cortices of Huntington's disease mice after intraperitoneal injection of compound 2.
- (21) FIG. 21. Effects of intraperitoneal injection of compound 2 on behavioral defects in Huntington's disease mice.
- (22) FIG. 22. Effects of compounds on the level of ATXN3 protein in fibroblasts of patients with spinocerebellar ataxia type 3.
- (23) FIG. 23. Effects of compounds on the level of mutant ATXN1 protein in HEK293T cells.

DETAILED DESCRIPTION

(24) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art. If there is a contradiction, the definition provided in this application shall prevail. When a trade name is present herein, it refers to the corresponding commodity or the active ingredient thereof. All patents, published patent applications and publications cited herein are incorporated herein by reference.

(25) General Terminology and Definitions

(26) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art. If there is a contradiction, the definition provided in this application shall prevail. When a trade name is present herein, it refers to the corresponding commodity or the active ingredient thereof. All patents, published patent applications and publications cited herein are incorporated herein by reference.

(27) The terms “including”, “comprising”, “having”, “containing”, or “relating to” and other variants thereof, as used herein, are inclusive or open-ended, and not exclusive of other elements or steps of methods that are not enumerated. Those skilled in the art should understand that the above terms such as “include” also cover the meaning of “consist of”.

(28) The term “one or more” or the similar expression “at least one” can mean, for example, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 or more.

(29) When the lower limit and the upper limit of a numerical range are disclosed, any numerical value and any included range falling within the range are specifically disclosed. In particular, each

value range of the values disclosed herein should be understood to mean each value and range covered in a wider range. For example, the expression “ATXN1 with polyQ length ≥ 40 ” can cover the case where the polyQ length ≥ 41 . For example, “ATXN2 with polyQ length ≥ 33 ” can cover the case where the polyQ length ≥ 34 . For example, “ATXN3 with polyQ length ≥ 41 ” can cover the case where the polyQ length ≥ 62 and can cover the case where the polyQ length is 74, for example. For example, “ATXN3 with polyQ length ≤ 41 ” can cover the case where the polyQ length is 27. For example, “ATXN7 with polyQ length ≥ 19 ” can cover the case where the polyQ length ≥ 38 . For example, “TBP with polyQ length ≥ 44 ” can cover the case where the polyQ length ≥ 45 . For example, “ATN1 with polyQ length ≥ 39 ” can cover the case where the polyQ length ≥ 49 . For example, “HTT with a polyQ length ≥ 36 ” can cover the case where the polyQ length is 47, 49, 55, 68, 72, 73, 111, 128, or 140. For another example, “HTT with polyQ length < 36 ” can cover the case where the polyQ length is 7, 16, 19, 23, or 25. For example, “AR with polyQ length ≥ 37 ” can cover the case where the polyQ length ≥ 38 .

(30) The expression m-n used as used herein refers to the range from m to n and the subrange composed of each point value and each point value. For example, the expression “C.sub.1-C.sub.8” or “C.sub.1-8” covers the range of 1-8 carbon atoms and should be understood to also cover any subrange and each point value, such as C.sub.2-C.sub.5, C.sub.3-C.sub.4, C.sub.1-C.sub.2, C.sub.1-C.sub.3, C.sub.1-C.sub.4, C.sub.1-C.sub.5, C.sub.1-C.sub.6, C.sub.1-C.sub.7, etc., and C.sub.1, C.sub.2, C.sub.3, C.sub.4, C.sub.5, C.sub.6, C.sub.7, C.sub.8, etc. For example, the expression “C.sub.3-C.sub.10” or “C.sub.3-10” should also be understood in a similar manner, for example, it can cover any subrange and point value contained therein, such as C.sub.3-C.sub.9, C.sub.6-C.sub.9, C.sub.6-C.sub.8, C.sub.6-C.sub.7, C.sub.7-C.sub.10, C.sub.7-C.sub.9, C.sub.7-C.sub.8, C.sub.8-C.sub.9, etc., and C.sub.3, C.sub.4, C.sub.5, C.sub.6, C.sub.7, C.sub.8, C.sub.9, C.sub.10, etc. For another example, the expression “3- to 10-membered” should be understood as covering any subrange and each point value, such as 3- to 5-membered, 3- to 6-membered, 3- to 7-membered, 3- to 8-membered, 4- to 5-membered, 4- to 6-membered, 4- to 7-membered, 4- to 8-membered, 5- to 7-membered, 5- to 8-membered, 6- to 7-membered, 6- to 8-membered, 9- to 10-membered, etc., and 3-, 4-, 5-, 6-, 7-, 8-, 9-, 10-membered, etc. Other similar expressions as used herein should also be understood in a similar way.

(31) The term “optional” or “optionally” means that the event or situation described later may or may not occur, and the description includes occurrence of said event or situation and non-occurrence of said event or situation.

(32) The terms “substitute” and “substituted” mean that one or more (e.g. one, two, three, or four) hydrogens are replaced by a selection from the indicated group, provided that not exceeding the normal valence of the designated atom under the circumstance and the substitution forms a stable compound. Combinations of substituents and/or variables are only allowed when the combination forms a stable compound. When describing that a substituent is absent, it should be understood that the substituent can be one or more hydrogen atoms, provided that the structure can result in a stable state of the compound.

(33) If a substituent is described as “optionally substituted”, the substituent may be unsubstituted or may be substituted. If an atom or group is described as being optionally substituted by one or more substituents on a list, one or more hydrogens on the atom or group can be optionally replaced by substituents which are selected independently. When the substituent is oxo (i.e., =O), it means that two hydrogen atoms are replaced.

(34) Unless otherwise specified, as used herein, the point of attachment of a substituent can be from any suitable position of the substituent. When a bond of a substituent is shown to pass through a bond connecting two atoms in a ring, such substituent can bond to any of the ring-forming atoms in the substitutable ring.

(35) When any variable (such as R), as well as variables with superscripts or subscripts (such as R.sup.X1, R.sup.X2, R.sup.7, R.sup.8, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a1, R.sup.b1, R.sup.c2,

etc.) appear more than once in the composition or structure of the compound, its definition in each case is independent at each occurrence. For example, if a group is substituted with 0, 1, 2, 3, or 4 R substituents, the group may optionally be substituted with up to four R substituents, and all the options of R substituent are all independent of each other.

(36) The term “halo” or “halogen” or “halogenated” should be understood to mean fluorine (F), chlorine (Cl), bromine (Br) or iodine (I) atoms, preferably fluorine, chlorine, bromine atoms.

(37) The term “alkyl” refers to a straight or branched saturated aliphatic hydrocarbon group composed of carbon atoms and hydrogen atoms, which is connected to the rest of the molecule by a single bond. “Alkyl” may have 1-8 atoms, referring to “C.sub.1-C.sub.8 alkyl”, for example, C.sub.1-4 alkyl, C.sub.1-3 alkyl, C.sub.1-2 alkyl, C.sub.3 alkyl, C.sub.4 alkyl, C.sub.1-6 alkyl, C.sub.3-6 alkyl. Non-limiting examples of alkyl groups include, but are not limited to methyl, ethyl, propyl, butyl, pentyl, hexyl, isopropyl, isobutyl, sec-butyl, tert-butyl, isopentyl, 2-methylbutyl, 1-methylbutyl 1-ethylpropyl, 1,2-dimethylpropyl, neopentyl, 1,1-dimethylpropyl, 4-methylpentyl, 3-methylpentyl, 2-methylpentyl, 1-methylpentyl, 2-ethylbutyl, 1-ethylbutyl, 3,3-dimethylbutyl, 2,2-dimethylbutyl, 1,1-dimethylbutyl, 2,3-dimethylbutyl, 1,3-dimethylbutyl or 1,2-dimethylbutyl, or their isomers.

(38) The term “alkylene”, when used alone or in combination with other groups herein, refers to a straight or branched saturated divalent hydrocarbon group. For example, the term “C.sub.1-6 alkylene” refers to an alkylene having 1-6 carbon atoms, for example, methylene, ethylene, propylene, butylene, pentylene, hexylidene, 1-methylethylene, 2-methylethylene, methylpropylene or ethylpropylene, etc.

(39) The term “alkenyl” refers to a straight or branched unsaturated aliphatic hydrocarbon group consisting of carbon atoms and hydrogen atoms with at least one double bond. The alkenyl group may have 2-8 carbon atoms, referring to, “C.sub.2-8 alkenyl”, for example, C.sub.2-alkenyl, C.sub.3-4 alkenyl. Non-limiting examples of alkenyl groups include, but are not limited to ethenyl, allyl, (E)-2-methylethenyl, (Z)-2-methylethenyl, (E)-but-2-enyl, (Z)-but-2-enyl, (E)-but-1-enyl, (Z)-but-1-enyl, etc.

(40) The term “alkynyl” refers to a straight or branched unsaturated aliphatic hydrocarbon group with at least one triple bond consisting of carbon atoms and hydrogen atoms. The alkynyl group may have 2-8 carbon atoms, referring to “C.sub.2-8 alkynyl”, for example, C.sub.2-4 alkynyl, C.sub.3-4 alkynyl. Non-limiting examples of alkynyl groups include, but are not limited to ethynyl, prop-1-ynyl, prop-2-ynyl, but-1-ynyl, but-2-ynyl, but-3-ynyl, etc.

(41) The term “cyclohydrocarbyl” refers to a saturated or unsaturated non-aromatic cyclic hydrocarbon group composed of carbon atoms and hydrogen atoms, preferably containing 1 or 2 rings. The said cyclohydrocarbyl may be a monocyclic ring, a fused polycyclic ring, a bridged ring or a spiro ring structure. The cyclohydrocarbyl group may have 3-10 carbon atoms, i.e. “C.sub.3-10 cyclohydrocarbyl”, for example, C.sub.3-8 cyclohydrocarbyl, C.sub.5 cyclohydrocarbyl, C.sub.6 cyclohydrocarbyl, C.sub.7 cyclohydrocarbyl. Non-limiting examples of cyclohydrocarbyl include, but are not limited to, cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, bicyclo[2.2.1]heptyl and spiro[3.3]heptyl, etc. The term also covers situations where the C atom can be substituted by oxo (=O).

(42) The term “heterocyclyl” or “heterocyclic hydrocarbon group” refers to a monocyclic or bicyclic ring system having, for example, 3-10 (suitably 3-8, more suitably 3-7, especially 4-6) ring atoms (3- to 10-membered, 3- to 8-membered, 3- to 7-membered, 3- to 6-membered), in which at least one ring atoms (for example, 1, 2 or 3) are heteroatoms selected from N, O and S, and the remaining ring atoms are C. The ring system can be saturated (also can be understood as the corresponding “heterocycloalkyl”) or unsaturated (i.e., having one or more double bonds and/or triple bonds in the ring). “Heterocyclyl” or “heterocyclic hydrocarbon group” does not possess aromaticity. The term also covers situations where the C atom can be substituted with oxo (=O) and/or the S atom on the ring can be substituted with 1 or 2 oxo (=O).

(43) The heterocyclyl may be, for example, a 4-membered ring, such as azetidiny, oxetanyl, or a 5-membered ring, such as tetrahydrofuranyl, dioxoliny, pyrrolidiny, imidazolidiny, pyrazolidiny, pyrroliny, oxopyrrolidiny, 2-oxoimidazolidin-1-yl; or 6-membered ring, such as tetrahydropyranyl, piperidiny, morpholiny, dithianyl, thiomorpholiny, piperaziny, 1,1-dioxo-1,2-thiazinan-2-yl or trithianyl; or a 7-membered ring, such as a diazepine ring. Optionally, the heterocyclyl can be benzo-fused.

(44) The heterocyclyl may be bicyclic without limitation, for example, a 5-membered fused 5-membered ring, such as hexahydrocyclopentane[c]pyrrole-2(1H)-yl; or a 5-membered fused 6-membered bicyclic ring, such as hexahydropyrrolo[1,2-a]pyrazine-2(1H)-yl.

(45) As mentioned above, the heterocycle may be unsaturated; that is to say, it may contain one or more double bonds without limitation. For example, an unsaturated heterocycle containing a nitrogen atom may be 1,6-dihydropyrimidine, 1,2-dihydropyrimidine, 1,4-dihydropyrimidine, 1,6-dihydropyridine, 1,2-dihydropyridine, 1,4-dihydropyridine, 2,3-dihydro-1H-pyrrole, 3,4-dihydro-1H-pyrrole, 2,5-dihydro-1H-pyrrolyl, 4H-[1,3,4]thiadiaziny, 4,5-dihydrooxazolyl or 4H-[1,4]thiaziny ring. An unsaturated heterocycle containing an oxygen atom may be 2H-pyran, 4H-pyran, or 2,3-dihydrofuran, and the unsaturated heterocycle containing a sulfur atom may be 2H-thiopyran or 4H-thiopyran. The heterocycle can be benzo-fused, but not limited thereto, such as dihydroisoquinoliny ring.

(46) The term “aryl” refers to an all-carbon monocyclic or fused polycyclic (such as bicyclic) aromatic ring group with a conjugated π -electron system. For example, aryl may have 6-14 carbon atoms, suitably 6-10, more suitably 6 or 10. Examples of aryl include, but are not limited to, phenyl, naphthyl, anthracenyl, etc.

(47) The term “heteroaryl” should be understood to preferably mean a monovalent monocyclic, bicyclic, or tricyclic aromatic ring system having 5, 6, 7, 8, 9 or 10 ring atoms (“5- to 10-membered heteroaryl”), especially 5 or 6 or 9 or 10 ring atoms, and at least one (suitably 1-4, more suitably 1, 2 or 3) of the ring atoms are heteroatoms such as oxygen, nitrogen, or sulfur. The heteroatoms may be the same with or different from each other. In addition, the heteroaryl can be benzo-fused in each case. In particular, the heteroaryl is selected from the group consisting of thienyl, furyl, pyrrolyl, oxazolyl, thiazolyl, imidazolyl, pyrazolyl, isoxazolyl, isothiazolyl, oxadiazolyl, triazolyl, thiadiazolyl, etc., and their benzo derivatives, such as benzofuranyl, benzothienyl, benzoxazolyl, benzisoxazolyl, benzimidazolyl, benzotriazolyl, indazole, indolyl, isoindolyl, etc.; or pyridyl, pyridaziny, pyrimidiny, pyraziny, triaziny, etc., and their benzo-fused derivatives, such as quinoliny, quinazoliny, isoquinoliny, etc., or azociny, indoliziny, puriny, etc., and their benzo derivatives; or cinnoliny, phthalaziny, quinazoliny, quinoxaliny, naphthyridiny, carbazolyl, acridiny, etc.

(48) The term “C.sub.1-C.sub.6 hydrocarbon chain” refers to a chain-like group composed of carbon atoms and hydrogen atoms, which may be straight or branched, and contains 1-8 (especially 1-5, such as 1, 2, 3, 4 or 5) carbon atoms. The hydrocarbon chain can be saturated (i.e., C.sub.1-C.sub.6 alkylene) or unsaturated; that is to say, it may contain one or more (preferably one) carbon-carbon double bond or triple bond.

(49) The alkylene may have 1-8 carbon atoms, referring to “C.sub.1-6 alkylene”, such as C.sub.1-5 alkylene, C.sub.1-4 alkylene, C.sub.1-3 alkylene, C.sub.1-2 alkylene, C.sub.3 alkylene, and C.sub.1 alkylene, i.e., methylene. Non-limiting examples of alkylene include but are not limited to methylene (—CH₂—), 1,1-ethylene (—CH(CH₃)—), 1,2-ethylene (—CH₂CH₂—), 1,1-propylene (—CH(CH₂CH₃)—), 1,2-propylene (—CH₂CH(CH₃)—), 1,3-propylene (—CH₂CH₂CH₂—), 1,4-butylen (—CH₂CH₂CH₂CH₂—) etc.

(50) The term “neurodegenerative disorder” refers to a disease caused by the loss or pathological change of neurons and/or their myelin sheaths. Characteristic pathological structures, such as insoluble aggregates of protein, can be observed in the brain neurons of patients with

neurodegenerative disorders. Insoluble aggregates may produce cytotoxicity, further leading to neuron loss and disease.

(51) The term “polyQ” or “polyglutamine” refers to the polyglutamine tract contained by the protein. Glutamine is encoded by cytosine-adenine-guanine (CAG) in the gene. The length of the polyglutamine is related to the number of CAG repeats in gene exons. Therefore, the increase in the number of CAG repeats in gene exons result in polyglutamine expansions in the synthesized protein. Proteins with abnormally expanded polyQ are known to be associated with some neurodegenerative disorder. As used herein, the gene name can use the form of “Q+number” to indicate the number of CAG repeats in exons, such as Q25 or Q72, which respectively indicate 25 repeats or 72 repeats of CAG in exons. In the protein name, the length of the polyglutamine can be expressed in the form of “Q+number” as above, such as Q27 or Q73, which means that the length of the polyglutamine is 27 Q (glutamine) or 73 Q, respectively. The CAG repetitions or glutamine repetitions indicated in the form of “Q+number” herein are all continuous repetitions. Unless otherwise specified, the length of polyQ as used herein refers to the length of the continuous polyglutamine.

(52) The term “polyQ-related neurodegenerative disorder” refers to a neurodegenerative disorder associated with abnormal expansion of polyQ, or a neurodegenerative disorder that responds to levels of proteins containing expanded polyQ. Neurodegenerative disorders are a group of disorders with clinical and genetic heterogeneity.

(53) “Normal polyQ” refers to the polyQ of a protein in a normal physiological state that has a length less than a specific number. Correspondingly, “abnormally expanded” means that the polyQ of the protein has a length longer than the normal length. For diseases or pathological conditions, the length of polyQ is longer. As an example, polyQ-related neurodegenerative disorders include but are not limited to spinocerebellar ataxia (SCA) type 1 (polyQ length ≥ 41), type 2 (polyQ length ≥ 34), type 3 (polyQ length ≥ 62), type 7 (polyQ length ≥ 38), type 12 (polyQ length ≥ 46), type 17 (polyQ length ≥ 45); and dentatorubral-pallidoluysian atrophy (DRPLA, polyQ length ≥ 49), Huntington's disease (HD, polyQ length ≥ 36) and spinal-bulbar muscular atrophy (SBMA, polyQ length ≥ 38). These diseases are caused by the expansion of CAG repeat regions on ATXN1, ATXN2, ATXN3, ATXN7, ATXN12, TBP, ATN1, HTT and AR genes, respectively (Lesley Jones et al., DNA repair in the trinucleotide repeat disorders, *Lancet Neurol.* 2017; 16: 88-96). Among them, spinocerebellar ataxia type 3 (SCA3, also known as Machado-Joseph disease, MJD) is the most common autosomal dominant spinocerebellar ataxia and is second only to HD as a common polyQ-related disorder in the world. SCA3 is caused by the abnormally expanded polyQ at the C-terminus of the coded protein ATXN3 resulting from the increase in the number of CAG repeats of the Ataxin-3 gene (ATXN3; also known as the MJD1 gene). Examples of normal polyQ proteins described herein include, but are not limited to, ATXN1 with polyQ length < 40 , ATXN2 with polyQ length < 33 , ATXN3 with polyQ length < 41 , ATXN7 with polyQ length < 19 , ATXN12 with polyQ length < 46 , TBP with polyQ length < 44 , ATN1 with polyQ length < 39 , HTT with polyQ length < 36 , and AR with polyQ length < 37 . Correspondingly, examples of the proteins with abnormally expanded polyQ described herein include, but are not limited to, ATXN1 with polyQ length ≥ 40 , ATXN2 with polyQ length ≥ 33 , ATXN3 with polyQ length ≥ 41 , ATXN7 with polyQ length ≥ 19 , and ATXN12 with polyQ length ≥ 46 , TBP with polyQ length ≥ 44 , ATN1 with polyQ length ≥ 39 , HTT with polyQ length ≥ 36 , and AR with polyQ length ≥ 37 .

(54) The term “pharmaceutically acceptable” refers to that when contacted with the patient's tissue within the scope of normal medical judgment, no undue toxicity, irritation, allergic reactions, etc. shall arise, having reasonable advantage-disadvantage ratios and effective for the intended use.

(55) The pharmaceutically acceptable salts of the compound of the present invention include acid addition salts and base addition salts thereof. Suitable acid addition salts are formed from acids that form pharmaceutically acceptable salts. Examples include hydrochloride, acetate, aspartate, benzoate, bicarbonate/carbonate, glucoheptonate, gluconate, nitrate, orotate, palmitic acid salt and

other similar salt. Suitable base addition salts are formed from bases that form pharmaceutically acceptable salts. Examples include aluminum salts, arginine salts, choline salts, magnesium salts, and other similar salts. The method for preparing the pharmaceutically acceptable salt of the compound of the present invention is known to those skilled in the art.

(56) The compound of the present invention may exist in specific geometric or stereoisomeric forms. The present invention contemplates all such compounds, including cis and trans isomers, (-)- and (+)-enantiomers, (R)- and (S)-enantiomers, diastereomers, (D)-isomers, (L)-isomers, and their racemic mixtures and other mixtures, for example, mixtures enriched in enantiomers or diastereomers, all of which are within the scope of the present invention. There may be other asymmetric carbon atoms in substituents such as alkyl. All these isomers and their mixtures are included in the scope of the present invention. In some embodiments, the preferred compounds are those isomeric compounds that show better biological activity. Purified or partially purified isomers and stereoisomers, or racemic mixtures or diastereomeric mixtures of the compound of the present invention are also included in the scope of the present invention. The purification and separation of such substances can be achieved by standard techniques known in the art.

(57) Optically pure enantiomers can be obtained by resolving racemic mixtures according to conventional methods, for example, by using optically active acids or bases to form diastereomeric salts, or by forming covalent diastereomers. A mixture of diastereomers can be separated into single diastereomers based on their physical and/or chemical differences by methods known in the art (for example, by chromatography or fractional crystallization). Then, release the optically active enantiomeric base or acid from the separated diastereomeric salt. Another method for separating racemic enantiomers can use chiral chromatography (such as a chiral HPLC column). The separated chiral isomers can be subjected to conventional derivatization or non-derivatization before separation, depending on what method can achieve more effective separation of chiral isomers. Enzymatic methods can also be used to separate derivatized or underivatized chiral isomers. Similarly, optically active raw materials can be used to obtain the optically pure compound of the present invention through chiral synthesis.

(58) In addition, the compound of the present invention may exist in the form of tautomers. The present invention includes all possible tautomers of the compound of the present invention, and also includes single tautomers or the form of any mixtures of said tautomers in any ratio.

(59) The compound of the present invention may exist in the form of solvates (preferably hydrates), wherein the compound of the present invention contains a polar solvent as a structural element of the compound crystal lattice, especially for example water, methanol, or ethanol. The amount of polar solvents, especially water, can be present in stoichiometric or non-stoichiometric ratios.

(60) The present invention also covers all possible crystalline forms or polymorphs of the compound of the present invention, which can be a single polymorph or a mixture of more than one polymorph in any ratio.

(61) The present invention also contemplates all pharmaceutically acceptable isotopically-labeled compounds, which are identical to the compounds of the invention except that one or more atoms are replaced by the atom(s) of the same atomic number but having atomic mass or mass number different from the atomic mass or mass number prevailing in nature.

(62) The metabolites of the compound of the present invention are also included within the scope of the present invention, namely the substances formed in the body when the compound of the present invention is administered. The metabolites of compounds can be identified by techniques known in the art, and their activity can be characterized by experimental methods. Such products can be produced, for example, by oxidation, reduction, hydrolysis, amidation, deamidation, esterification, enzymatic hydrolysis, etc. of the administered compound. Therefore, the present invention includes metabolites of the compound of the present invention, including compounds prepared by contacting the compound of the present invention with a mammal for a time sufficient to produce its metabolites.

(63) The present invention further includes within its scope the prodrugs of the compound of the present invention, which are certain derivatives of the compound of the present invention that have less or no pharmacological activity themselves but when administered to or on the body, can be converted into the compound of the present invention with the desired activity by, for example, hydrolytic cleavage. Usually, such prodrugs will be functional group derivatives of the compound, which are easily converted into the desired therapeutically active compound in vivo. For a review of prodrugs and their preparation methods, see, for example, J. Rautio et al. *Nature Reviews Drug Discovery* (2008) 7, 255-270 and *Prodrugs: Challenges and Rewards* (V. Stella et al. ed. Springer, 2007). The prodrugs of the present invention can be prepared, for example, by replacing an appropriate functional group in the compound of the present invention with a certain moiety known to those skilled in the art as “pro-moiety”.

(64) The term “polymorphism” or “polymorph” refers to a single polymorph or a mixture of more than one polymorph in any ratio.

(65) The term “crystal form” or “crystal” refers to any solid substance exhibiting a three-dimensional order, as opposed to an amorphous solid substance, which produces a characteristic X-ray powder diffraction pattern with sharp and defined peaks.

(66) The term “amorphous” refers to any solid substance which lacks order in three dimensions.

(67) The term “hydrate” describes a solvate containing a drug and a stoichiometric or non-stoichiometric amount of water.

(68) The term “pharmaceutically acceptable carrier” refers to those substances that have no obvious stimulating effect on the organism and will not damage the biological activity and performance of the active compound. “Pharmaceutically acceptable carriers” include but are not limited to glidants, sweeteners, diluents, preservatives, dyes/colorants, flavors, surfactants, wetting agents, dispersants, disintegrants, stabilizer, solvent or emulsifier. Non-limiting examples of the carrier include calcium carbonate, calcium phosphate, various sugars and various starches, cellulose derivatives, gelatin, vegetable oils and polyethylene glycols, etc.

(69) The term “administration” or “administering” or the like refers to a method that enables a compound or composition to be delivered to a desired site of biological action. Such methods comprise but not limited to oral or parenteral (including intracerebroventricular, intravenous, subcutaneous, intraperitoneal, intramuscular, intravascular injection or infusion), local, rectal administration or the like. Especially injection or oral.

(70) As used herein, The terms “treat,” “treating” or “treatment,” and other grammatical equivalents as used herein, include alleviating, abating or ameliorating a disease or condition, preventing additional symptoms, ameliorating or preventing the underlying metabolic causes of symptoms, inhibiting the disease or condition, e.g., arresting the development of the disease or condition, relieving the disease or condition, causing regression of the disease or condition, relieving a condition caused by the disease or condition, or stopping the symptoms of the disease or condition, and are intended to include prophylaxis. The terms further include achieving a therapeutic benefit and/or a prophylactic benefit. By therapeutic benefit is meant eradication or amelioration of the underlying disorder being treated. Also, a therapeutic benefit is achieved with the eradication or amelioration of one or more of the physiological symptoms associated with the underlying disorder such that an improvement is observed in the patient, notwithstanding that the patient may still be afflicted with the underlying disorder. For prophylactic benefit, the compositions may be administered to a patient at risk of developing a particular disease, or to a patient reporting one or more of the physiological symptoms of a disease, even though a diagnosis of this disease may not have been made.

(71) The terms “active ingredient”, “therapeutic agent”, “active substance” or “active agent” refer to a chemical entity that can effectively treat or prevent the target disorder, disease, or symptom.

(72) For drugs, drug units or active ingredients, the terms “effective amount”, “therapeutically effective amount” or “prophylactically effective amount” refer to an amount of a drug or agent that

has acceptable side effects but is sufficient to achieve the desired effect. The effective amount may be determined individually and depends on the age and general condition of the receptor as well as specific active substance. The effective amount in specific case can be determined by a person skilled in the art through conventional test.

(73) As used herein, the term “subject” includes a human or non-human animal. An exemplary human subject includes a human subject having a disease (such as one described herein) (referred to as a patient), or a normal subject. The term “non-human animal” as used herein includes all vertebrates, such as non-mammals (e.g. birds, amphibians, reptiles) and mammals, such as non-human primates, livestock and/or domesticated animals (such as sheep, dog, cat, cow, pig and the like).

(74) The following detailed description of the invention is intended to illustrate non-limiting embodiments, so that others skilled in the art can more fully understand the technical solution of the present invention, its principles and practical applications, so that others skilled in the art can modify and implement the present invention in various manners so that it can be optimally adapted to the requirements of specific applications.

(75) The Compound of the Present Invention and Use Thereof

(76) In one aspect, the present invention provides use of a compound of formula (I), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder

(77) ##STR00006## wherein: ring A is a benzene ring; ring B is a saturated or unsaturated 5- or 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S. ring C is selected from C.sub.6-10 aryl and 5- to 10-membered heteroaryl, wherein the aryl or heteroaryl is optionally substituted with one or more substituents each independently selected from R.sup.X1.

(78) L.sup.1 is a bond, or a C.sub.1-6 hydrocarbon chain; or ring C is absent, and L.sup.1 is absent; R.sup.1 is =Y, wherein Y is O or S, or OR.sup.7; at each occurrence, R.sup.2 is each independently selected from hydrogen, halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, =O, =S, =NR.sup.a1, —OR.sup.a1, —SR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, —C(=O)NR.sup.a1R.sup.b1, C(=O)R.sup.a1, —S(=O).sub.2OR.sup.a1, —S(=O).sub.2R.sup.a1, —S(=O).sub.2NR.sup.a1R.sup.b1, —S(=O)R.sup.a1, —C(=S)OR.sup.a1, —C(=S)NR.sup.a1R.sup.b1, C(=S)R.sup.a1, P(=O)(OR.sup.a1)OR.sup.b1, —C(=NR.sup.a1)NR.sup.b1R.sup.c1, —OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, =O, =S, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2, C(=O)R.sup.a2, S(=O).sub.2OR.sup.a2, —S(=O).sub.2R.sup.a2, —S(=O).sub.2NR.sup.a2R.sup.b2, —S(=O)R.sup.a2 and —C(=NR.sup.a2)NR.sup.b2R.sup.c2; R.sup.3, R.sup.4, R.sup.5, R.sup.6 are each independently selected from H and R.sup.X2; at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, —OR.sup.7, —SR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —C(=O)R.sup.7, —S(=O).sub.2OR.sup.7, —S(=O).sub.2R.sup.7, —S(=O).sub.2NR.sup.7R.sup.8, —OS(=O).sub.2R.sup.7, —NS(=O).sub.2R.sup.7R.sup.8 and —S(=O)R.sup.7, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6

cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C.sub.1-6 alkyl), —S(C.sub.3-6 cyclohydrocarbyl), —S(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, =O, —COOH and C.sub.1-6 alkyl; at each occurrence, R.sup.7, R.sup.8 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.14 alkyl and C.sub.6-10 aryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl or aryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl); at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6-alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.14 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, —OR.sup.Y1, —SR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2, —C(=O)R.sup.Y1, —S(=O).sub.2OR.sup.Y1, —S(=O).sub.2R.sup.Y1, —S(=O).sub.2NR.sup.Y1R.sup.Y2 and —S(=O)R.sup.Y1, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents independently selected from halogen, =O, =S, —OR.sup.Y3, —SR.sup.Y3, —NR.sup.Y3R.sup.Y4, —C(=O)R.sup.Y3, —C(=O)OR.sup.Y3 and —C(=O)NR.sup.Y3R.sup.Y4; at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-8 alkyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, —OH, —SH, —NH.sub.2, =O and —COOH; n is 1 or 2; provided that the compound of formula (I) does not include the following structure:

(79) ##STR00007## wherein R.sup.3 is selected from —O(C.sub.1-6 alkyl) and —O(C.sub.1-4 alkylene-C.sub.6-10 aryl); R.sup.10 is H; and the compound of formula (I) does not include the following structure:

(80) ##STR00008## wherein R.sup.4 is —O(C.sub.1-6 alkyl); R.sup.5 is halogen; R.sup.10 is CF.sub.3; and the compound of formula (I) does not include the following structure:

(81) ##STR00009## wherein R.sup.4 is selected from —O(C.sub.1-6 alkyl) and —O(C.sub.1-4 alkylene-C.sub.6-10 aryl); ring C is C.sub.6-10 aryl, which is optionally substituted with one or more substituents each independently selected from R.sup.X1; and the compound of formula (I) does not include the following structure:

(82) ##STR00010## wherein R.sup.4 is selected from —OH and —O(C.sub.1-6 alkyl); R.sup.5 is halogen; ring C is 5- to 10-membered heteroaryl, which is optionally substituted with one or more substituents each independently selected from R.sup.X1.

(83) In one embodiment, R.sup.1 is =O. In another embodiment, R.sup.1 is =S. In another

embodiment, R.sup.1 is OR.sup.7.

(84) In one embodiment, ring B is a saturated or unsaturated 5-membered or 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S. In one embodiment, ring B is a saturated or unsaturated 5-membered heterocycle, wherein the heterocycle contains 1 or 2 heteroatoms each independently selected from N and O. In another embodiment, ring B is dihydropyrrole. In another embodiment, ring B is selected from 2,3-dihydro-1H-pyrrole and 3,4-dihydro-1H-pyrrole, preferably 2,3-dihydro-1H-pyrrole. In yet another embodiment, ring B is pyrrolidine.

(85) ##STR00011##

(86) In a more preferred embodiment, the A-B ring system is wherein Y is O or S; and ring C is a 5- to 7-membered heteroaryl, preferably a 5- to 6-membered heteroaryl, especially a 5-membered heteroaryl, wherein the heteroaryl is optionally substituted with 1, 2, 3, 4 or 5 substituents each independently selected from R.sup.X1. In a particular embodiment, the A-B ring system is

(87) ##STR00012##

and ring C is a 5- to 7-membered heteroaryl, preferably a 5- to 6-membered heteroaryl, especially a 5-membered heteroaryl, the heteroaryl is optionally substituted by 1, 2, 3, 4 or 5 substituents each independently selected from R.sup.X1. In another embodiment, ring C contains 1, 2, 3 or 4 heteroatoms, each of which is independently selected from N, O and S, preferably selected from N and O. In yet another embodiment, ring C contains at least one N atom. In one embodiment, ring C is a 5-membered heteroaryl group containing 1 or 2 N atoms, optionally substituted with 1, 2, 3, 4, or 5 substituents each independently selected from R.sup.X1. In another embodiment, ring C is selected from pyrrole and imidazole.

(88) In one embodiment, ring B is a saturated or unsaturated 6-membered heterocycle wherein the heterocycle contains 1 or 2 heteroatoms each independently selected from N and O. In another embodiment, ring B is dihydropyrimidine. In a preferred embodiment, ring B is selected from 1,6-dihydropyrimidine, 1,2-dihydropyrimidine and 1,4-dihydropyrimidine.

(89) ##STR00013##

(90) In a more preferred embodiment, the A-B ring system is wherein Y is O or S. In a particular embodiment, the A-B ring system is

(91) ##STR00014##

(92) In yet another embodiment, ring B is 2H-pyran or 4H-pyran. In a preferred embodiment, the A-B ring system is

(93) ##STR00015##

wherein Y is O or S. In a particular embodiment, the A-B ring system is

(94) ##STR00016##

(95) In yet another embodiment, ring C is phenyl, optionally substituted with 1, 2, 3, 4, or 5 substituents each independently selected from R.sup.X1.

(96) In one embodiment, L.sup.1 is a bond. In another embodiment, L.sup.1 is a C.sub.1-C.sub.6 hydrocarbon chain.

(97) In a preferred embodiment, L.sup.1 is a C.sub.1-C.sub.2 hydrocarbon chain.

(98) In one embodiment, the A-B ring system is

(99) ##STR00017##

ring C is absent, and L.sup.1 is absent.

(100) In another embodiment: ring A is a benzene ring; ring B is a saturated or unsaturated 5-membered or 6-membered heterocycle, wherein heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S; ring C is selected from C.sub.6-10 aryl and 5- to 10-membered heteroaryl, wherein aryl or heteroaryl is optionally substituted with one or more substituents each independently selected from R.sup.X1; L.sup.1 is a bond, or a C.sub.1-6 hydrocarbon chain; R.sup.1 is =Y, wherein Y is O or S, or OR.sup.7; at each occurrence, R.sup.2 is each independently selected from hydrogen, halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-

8 alkenyl, C.sub.2-8 alkynyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, =O, =S, =NR.sup.a1, —OR.sup.a1, —SR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, —C(=O)NR.sup.a1R.sup.b1, —C(=O)R.sup.a1, —S(=O).sub.2OR.sup.a1, —S(=O).sub.2R.sup.a1, —S(=O).sub.2NR.sup.a1R.sup.b1, —S(=O)R.sup.a1, —C(=S)OR.sup.a1, —C(=S)NR.sup.a1R.sup.b1, —C(=S)R.sup.a1, —P(=O)(OR.sup.a1)OR.sup.b1, —C(=NR.sup.a1)NR.sup.b1R.sup.c1, —OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents independently selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, =O, =S, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2, —C(=O)R.sup.a2, —S(=O).sub.2OR.sup.a2, —S(=O).sub.2R.sup.a2, —S(=O).sub.2NR.sup.a2R.sup.b2, —S(=O)R.sup.a2 and —C(=NR.sup.a2)NR.sup.b2R.sup.c2; R.sup.3, R.sup.4, R.sup.5, R.sup.6 are each independently selected from H and R.sup.X2; at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, —OR.sup.7, —SR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —C(=O)R.sup.7, —S(=O).sub.2OR.sup.7, —S(=O).sub.2R.sup.7, —S(=O).sub.2NR.sup.7R.sup.8, —OS(=O).sub.2R.sup.7, —NS(=O).sub.2R.sup.7R.sup.8 and —S(=O)R.sup.7, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C.sub.1-6 alkyl), —S(C.sub.3-6 cyclohydrocarbyl), —S(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), alkylene-3- to 7-membered heterocyclyl).sub.2, =O, —COOH and C.sub.1-6 alkyl; at each occurrence, R.sup.7, R.sup.8 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl and C.sub.6-10 aryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl or aryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl); at each occurrence, R.sup.a1, R.sup.b1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, —OR.sup.Y1, —SR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2, —C(=O)R.sup.Y1, —S(=O).sub.2OR.sup.Y1, —S(=O).sub.2R.sup.Y1, —S(=O).sub.2NR.sup.Y1R.sup.Y2 and —S(=O)R.sup.Y1, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, =O, =S, —OR.sup.Y3, —SR.sup.Y3, —NR.sup.Y3R.sup.Y4, —C(=O)R.sup.Y3, —C(=O)OR.sup.Y3 and —C(=O)NR.sup.Y3R.sup.Y4;

at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-8 alkyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, —OH, —SH, —NH.sub.2, =O and —COOH; n is 1; and provided that the compound of formula (I) does not include the following structure:

(101) ##STR00018## wherein R.sup.4 is selected from —O(C.sub.1-6 alkyl) and —O(C.sub.1-4 alkylene-C.sub.6-10 aryl); ring C is C.sub.6-10 aryl, which is optionally substituted with one or more substituents each independently selected from R.sup.X1; and the compound of formula (I) does not include the following structure:

(102) ##STR00019## wherein R.sup.4 is selected from —OH and —O(C.sub.1-6 alkyl); R.sup.5 is halogen; ring C is 5- to 10-membered heteroaryl, which is optionally substituted with one or more substituents each independently selected from R.sup.X1.

(103) The structure described in the following formula (II) falls within the scope of the structure described in formula (I). In another aspect of the present invention, provided is use of a compound of formula (II), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder

(104) ##STR00020## wherein:

(105) Y is O or S; ring C is 5- to 7-membered heteroaryl, the heteroaryl is optionally substituted with one or more substituents each independently selected from R.sup.X1; R.sup.2 is selected from H and C.sub.1-8 alkyl; L.sup.1 is a bond, or a C.sub.1-C.sub.6 hydrocarbon chain; R.sup.3, R.sup.4, R.sup.5, R.sup.6 are each independently selected from H and R.sup.X2; wherein R.sup.X1, R.sup.X2 are as defined in formula (I).

(106) In one embodiment, Y is O. When Y is O, the formula (II) is

(107) ##STR00021##

(108) In one embodiment, ring C is a 5- to 6-membered heteroaryl, wherein the heteroaryl is optionally substituted with 1, 2, 3, 4 or 5 substituents each independently selected from R.sup.X1. In a preferred embodiment, ring C is a 5-membered heteroaryl, which is optionally substituted with 1, 2, 3, 4 or 5 substituents each independently selected from R.sup.X1. In another embodiment, ring C contains 1, 2, 3 or 4 heteroatoms, each of which is independently selected from N, O and S, preferably selected from N and O. In another embodiment, ring C contains at least one N atom. In one embodiment, ring C is a 5-membered heteroaryl containing 1 or 2 N atoms, optionally substituted with 1, 2, 3, 4, or 5 substituents each independently selected from R.sup.X1. In another embodiment, ring C is selected from pyrrole and imidazole.

(109) In one embodiment, L.sup.1 is a bond. In another embodiment, L.sup.1 is a C.sub.1-C.sub.6 hydrocarbon chain. In a preferred embodiment, L.sup.1 is a C.sub.1-C.sub.2 hydrocarbon chain. In one embodiment, L.sup.1 is methylene or methenyl. In a special embodiment, L.sup.1 is methenyl. In another particular embodiment, L.sup.1 is

(110) ##STR00022##

(111) In another particular embodiment, the compound of formula (II) is selected from:

(112) ##STR00023##

(113) The structure described in the following formula (III) falls within the scope of the structure described in formula (I). In another aspect of the present invention, provided is use of a compound of formula (III), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder wherein:

(114) ##STR00024## ring B is a saturated or unsaturated 6-membered heterocycle, wherein the heterocycle contains 1, 2, or 3 heteroatoms each independently selected from N, O and S; Ring C is a C.sub.6-10 aryl, optionally substituted with one or more substituents each independently selected from R.sup.X1; or ring C is absent, and L.sup.1 is absent; ring A, L.sup.1, R.sup.1, R.sup.2, R.sup.3, R.sup.4, R.sup.5, R.sup.6, R.sup.X1, n are as defined in formula (I); provided that the compound of formula (III) does not include the following structure:

(115) ##STR00025## wherein R.sup.3 is selected from —O(C.sub.1-6 alkyl) and —O(C.sub.1-4 alkylene-C.sub.6-10 aryl); R.sup.10 is H; and the compound of formula (III) does not include the following structure:

(116) ##STR00026## wherein R.sup.4 is —O(C.sub.1-6 alkyl); R.sup.5 is halogen; R.sup.10 is CF.sub.3; and the compound of formula (III) does not include the following structure:

(117) ##STR00027## wherein R.sup.4 is selected from —O(C.sub.1-6 alkyl) and —O(C.sub.1-4 alkylene-C.sub.6-10 aryl); ring C is a C.sub.6-10 aryl, which is optionally substituted with one or more substituents each independently selected from R.sup.X1; and the compound of formula (III) does not include the following structure:

(118) ##STR00028## wherein R.sup.4 is selected from —OH and —O(C.sub.1-6 alkyl); R.sup.5 is halogen; ring C is a 5- to 10-membered heteroaryl, wherein heteroaryl is substituted with one or more substituents each independently selected from R.sup.X1.

(119) In one embodiment, ring B is a saturated or unsaturated 6-membered heterocycle, wherein heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S. In one embodiment, ring B is a saturated or unsaturated 6-membered heterocycle, wherein heterocycle contains 1 or 2 heteroatoms each independently selected from N and O. In another embodiment, ring B is dihydropyrimidine. In a preferred embodiment, ring B is selected from 1,6-dihydropyrimidine, 1,2-dihydropyrimidine and 1,4-dihydropyrimidine.

(120) ##STR00029##

(121) In a more preferred embodiment, the A-B ring system is wherein Y is O or S. In a particular embodiment, the A-B ring system is

(122) ##STR00030##

(123) In yet another embodiment, ring B is 2H-pyran or 4H-pyran. In a preferred embodiment, the A-B ring system is

(124) ##STR00031##

wherein Y is O or S. In a particular embodiment, the A-B ring system is

(125) ##STR00032##

(126) In yet another embodiment, ring C is phenyl, optionally substituted with 1, 2, 3, 4, or 5 substituents each independently selected from R.sup.X1.

(127) In one embodiment, L.sup.1 is a bond. In another embodiment, L.sup.1 is a C.sub.1-C.sub.6 hydrocarbon chain.

(128) In a preferred embodiment, L.sup.1 is a C.sub.1-C.sub.2 hydrocarbon chain.

(129) In one embodiment, the A-B ring system is

(130) ##STR00033##

ring C is absent, and L.sup.1 is absent.

(131) In an alternative embodiment, ring B and ring C of the compound of formula (III) are further connected through L.sup.2 to obtain a variant of the compound of formula (III), which has the following structure of formula (III')

(132) ##STR00034## wherein: ring C is C.sub.6-10 aryl, which is optionally substituted with one or more substituents each independently selected from R.sup.X1; R.sup.1 is H, =O, or is OR.sup.7; L.sup.1 is a bond, or is a C.sub.1-C.sub.2 hydrocarbon chain; L.sup.2 is a bond, or is a C.sub.1-C.sub.2 hydrocarbon chain; provided that L.sup.1 and L.sup.2 are not bonds at the same time; R.sup.3, R.sup.4, R.sup.5, R.sup.6 are each independently selected from H and R.sup.X2; ring A, ring B, R.sup.2, n, R.sup.X1 are as defined in formula (III).

(133) In one embodiment, R.sup.1 is H.

(134) In another embodiment, R.sup.2 is —OH.

(135) In one embodiment, at each occurrence, R.sup.X2 is each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH and —O(C.sub.1-6 alkyl).

(136) The structure described in the following formula (IV) falls within the scope of the structure described in formula (I) and formula (III). In another aspect of the present invention, provided is use of a compound of formula (IV), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder

(137) ##STR00035## wherein: Y is O or S; X is O; R.sup.9 is selected from H, halogen, C.sub.1-6 alkyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, —OR.sup.a1, —SR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, —C(=O)NR.sup.a1R.sup.b1, —C(=O)R.sup.a1, —S(=O).sub.2OR.sup.a1, —S(=O).sub.2R.sup.a1, —S(=O).sub.2NR.sup.a1R.sup.b1 and —S(=O)R.sup.a1, wherein the alkyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2, —C(=O)R.sup.a2, —S(=O).sub.2OR.sup.a2, —S(=O).sub.2R.sup.a2, —S(=O).sub.2NR.sup.a2R.sup.b2 and —S(=O)R.sup.a2; wherein R.sup.a1, R.sup.b1, R.sup.a2, R.sup.b2 are as defined in formula (III); R.sup.10 is selected from H, halogen, C.sub.1-6 alkyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl; R.sup.3 is selected from H, halogen, C.sub.1-6 alkyl, —OH, —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2; R.sup.4 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.7, —SR.sup.7 and —NR.sup.7R.sup.8; at each occurrence, R.sup.7, R.sup.8 are each independently selected from H and C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl); wherein R.sup.7, R.sup.8 are as defined in formula (III); R.sup.5 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl); R.sup.6 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —O(benzyl), —SH, —S(C.sub.1-6 alkyl), —S(benzyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2 and —NH(benzyl), wherein the alkyl or benzyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2; provided that when R.sup.4 is —O(C.sub.1-6 alkyl) and R.sup.5 is halogen, R.sup.10 is not CF.sub.3.

(138) In one embodiment, Y is O. When Y is O, formula (IV) is

(139) ##STR00036##

(140) In one embodiment, R.sup.9 is selected from H, halogen, C.sub.1-6 alkyl, —OR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, C(=O)NR.sup.a1R.sup.b1 and —

S(=O).sub.2NR.sup.a1R.sup.b1, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2 and —C(=O)R.sup.a2. In one embodiment, R.sup.9 is selected from H, halogen, —OR.sub.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1 and —C(=O)NR.sup.a1R.sup.b1. In one embodiment, R.sup.9 is —OR.sup.a1 or —NR.sup.a1R.sup.b1. In another embodiment, R.sup.9 is —C(=O)OR.sup.a2 or —C(=O)NR.sup.a2R.sup.b2. In a particular embodiment, R.sup.9 is —NHC(=O)(C.sub.1-6 alkyl). In a particular embodiment, R.sup.9 is —NHC(=O)CH.sub.3. (141) In one embodiment, R.sup.10 is selected from H, halogen and methyl. In another embodiment, R.sup.10 is H.

(142) In one embodiment, at each occurrence, R.sup.a1, R.sup.b1 are each independently selected from H, C.sub.1-6 alkyl and —C(=O)R.sup.Y1, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OR.sup.Y3, —SR.sup.Y3, —NR.sup.Y3R.sup.Y4, —C(=O)R.sup.Y3, —C(=O)OR.sup.Y3 and —C(=O)NR.sup.Y3R.sup.Y4. Wherein R.sup.Y1, R.sup.Y3, R.sup.Y4 are as defined in formula (III). In one embodiment, at each occurrence, R.sup.a1R.sup.b1 are independently selected from H, C.sub.1-6 alkyl and —C(=O)(C.sub.1-6 alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —C(=O)(C.sub.1-6 alkyl), —C(=O)O(C.sub.1-6 alkyl), —C(=O)O(C.sub.3-6 cyclohydrocarbyl), —C(=O)NH(C.sub.1-6 alkyl) and —C(=O)N(C.sub.1-6 alkyl).sub.2. In one embodiment, at each occurrence, R.sup.a1, R.sup.b1 are independently selected from H and C.sub.1-6 alkyl.

(143) In one embodiment, at each occurrence, R.sup.a2, R.sup.b2 are each independently selected from H and C.sub.1-6 alkyl.

(144) In one embodiment, R.sup.3 is selected from H, —OH and C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2 and —COOH. In one embodiment, R.sup.3 is selected from H, halogen, methyl, —OH, —NH.sub.2 and —NHCH.sub.3, wherein the methyl is optionally substituted with substituents selected from halogen, —OH, —NH.sub.2 and —NH(C.sub.1-2 alkyl). In one embodiment, R.sup.3 is selected from H, halogen, methyl, —OH, —NH.sub.2 and —NHCH.sub.3, wherein the methyl is optionally substituted with substituents selected from halogen, —OH, —OCH.sub.3, —NH.sub.2 and —NHCH.sub.3. In one embodiment, R.sup.3 is selected from H, halogen, methyl, —OH and —NH.sub.2. In one embodiment, R.sup.3 is selected from H, F, Cl, methyl, —OH and —NH.sub.2. In another embodiment, R.sup.3 is selected from H, F, methyl, —OH and —NH.sub.2. In a particular embodiment, R.sup.3 is selected from H, methyl and —OH. In another particular embodiment, R.sup.3 is H. In another particular embodiment, R.sup.3 is —OH. In another embodiment, R.sup.3 is dimethylaminomethyl.

(145) In one embodiment, R.sup.4 is selected from H, —OR.sup.7, —SR.sup.7 and —NR.sup.7R.sup.8; at each occurrence, R.sup.7, R.sup.8 are each independently selected from H and C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl) and —C(=O)N(C.sub.1-6 alkyl).sub.2. In one embodiment, R.sup.4 is C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —COOH, —

C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl) and —C(=O)N(C.sub.1-6 alkyl).sub.2. In a particular embodiment, R.sup.4 is —CH.sub.2COOH. In another embodiment, R.sup.4 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2. In one embodiment, R.sup.4 is selected from H, halogen, C.sub.1-3 alkyl, —OH, —O(C.sub.1-3 alkyl), —NH.sub.2, —NH(C.sub.1-3 alkyl), —N(C.sub.1-3 alkyl).sub.2 and —COOH, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-2 alkyl), —NH.sub.2, —NH(C.sub.1-2 alkyl), —N(C.sub.1-2 alkyl).sub.2 and —COOH.

(146) In one embodiment, R.sup.5 is selected from H and C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C.sub.1-6 alkyl), —S(C.sub.3-6 cyclohydrocarbyl), —S(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2 and —COOH. In another embodiment, R.sup.5 is selected from H, halogen, C.sub.1-3 alkyl, —OH, —O(C.sub.1-3 alkyl), —NH.sub.2, —NH(C.sub.1-3 alkyl), —N(C.sub.1-3 alkyl).sub.2 and —COOH, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-2 alkyl), —NH.sub.2, —NH(C.sub.1-2 alkyl), —N(C.sub.1-2 alkyl).sub.2 and —COOH. In one embodiment, R.sup.5 is selected from H, halogen, C.sub.1-3 alkyl, —OH, —O(C.sub.1-3 alkyl), —NH.sub.2, —NH(C.sub.1-3 alkyl) and —N(C.sub.1-3 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-2 alkyl), —NH.sub.2, —NH(C.sub.1-2 alkyl), —N(C.sub.1-2 alkyl).sub.2. In one embodiment, R.sup.5 is dimethylaminomethyl.

(147) In one embodiment, R.sup.6 is selected from H, halogen, C.sub.1-6 alkyl, —OH and —NH.sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH and —NH.sub.2. In one embodiment, R.sup.6 is selected from H, F, Cl, Br, C.sub.1-6 alkyl, —OH and —NH.sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH and —NH.sub.2. In one embodiment, R.sup.6 is selected from H, F, Cl, Br, methyl, —OH and —NH.sub.2. In one embodiment, R.sup.6 is H or —OH.

(148) In one embodiment, R.sup.3 is methyl, and R.sup.4 is selected from —OH, —NH.sub.2, —NH(C.sub.1-3 alkyl), —N(C.sub.1-3 alkyl).sub.2, and, C.sub.1-3 alkyl substituted with —COOH, wherein the substituted C.sub.1-3 alkyl is substituted with one or more substituents selected from —OH, —NH.sub.2, —NH(C.sub.1-2 alkyl), —N(C.sub.1-2 alkyl).sub.2 and —COOH. In one embodiment, R.sup.3 is methyl, and R.sup.4 is selected from —NH.sub.2, —NH(C.sub.1-3 alkyl), —N(C.sub.1-3 alkyl).sub.2, and, C.sub.1-3 alkyl substituted with —COOH, wherein substituted C.sub.1-3 alkyl is substituted with one or more substituents selected from —OH, —NH.sub.2, —NH(C.sub.1-2 alkyl), —N(C.sub.1-2 alkyl).sub.2 and —COOH.

(149) In another embodiment, R.sup.3 is methyl, and R.sup.5 is selected from —OH, —NH.sub.2, —NH(C.sub.1-3 alkyl), —N(C.sub.1-3 alkyl).sub.2, and, C.sub.1-3 alkyl substituted with —COOH, wherein the substituted C.sub.1-3 alkyl is substituted with one or more substituents selected from —OH, —NH.sub.2, —NH(C.sub.1-2 alkyl), —N(C.sub.1-2 alkyl).sub.2 and —

COOH. In another embodiment, R³ is methyl, and R⁵ is selected from —NH₂, —NH(C₁₋₃ alkyl), —N(C₁₋₃ alkyl)₂, and, C₁₋₃ alkyl substituted with —COOH, wherein the substituted C₁₋₃ alkyl is substituted with one or more substituents selected from —OH, —NH₂, —NH(C₁₋₂ alkyl), —N(C₁₋₂ alkyl)₂ and —COOH.

(150) The structure described in the following formula (V) falls within the scope of the structure described in formula (I), formula (III) and formula (IV). In another aspect of the present invention, provided is use of a compound of formula (V), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder,

(151) ##STR00037## wherein: Y, R⁹, R¹⁰, R³, R⁴, R⁵, R⁶ are as defined in formula (IV); in one embodiment, Y is O. When Y is O, formula (V) is

(152) ##STR00038##

(153) In a particular embodiment, compound of formula (V) is

(154) ##STR00039##

(155) The structure described in the following formula (VI) falls within the scope of the structure described in formula (I) and formula (III). In another aspect of the present invention, there is provided use of a compound of formula (VI), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder

(156) ##STR00040## wherein: ring B is a saturated or unsaturated 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S ring C is a C₆₋₁₀ aryl, wherein the aryl optionally substituted with one or more substituents each independently selected from R^{X1}; L¹ is a bond, or a C₁₋₆ hydrocarbon chain; R² is selected from H, halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, C₃₋₁₀ cyclohydrocarbyl, C₃₋₁₀ cyclohydrocarbyl-C₁₋₄ alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C₁₋₄ alkyl, C₆₋₁₀ aryl-C₁₋₄ alkyl, 5- to 10-membered heteroaryl-C₁₋₄ alkyl, =O, =S, =NR^{a1}, —OR^{a1}, —SR^{a1}, —NR^{a1}R^{b1}, —C(=O)OR^{a1}, —C(=O)NR^{a1}R^{b1}, —C(=O)R^{a1}, —S(=O)₂OR^{a1}, —S(=O)₂R^{a1}, —S(=O)₂NR^{a1}R^{b1}, —S(=O)R^{a1}, —C(=S)OR^{a1}, —C(=S)NR^{a1}R^{b1}, —C(=S)R^{a1}, —P(=O)(OR^{a1})OR^{b1}, —C(=NR^{a1})NR^{b1}R^{c1}, —OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₂₋₈ alkynyl, =O, =S, —OR^{a2}, —SR^{a2}, —NR^{a2}R^{b2}, —C(=O)OR^{a2}, —C(=O)NR^{a2}R^{b2}, —C(=O)R^{a2}, —S(=O)₂OR^{a2}, —S(=O)₂R^{a2}, —S(=O)₂NR^{a2}R^{b2}, —S(=O)R^{a2} and —C(=NR^{a2})NR^{b2}R^{c2}; R³, R⁴, R⁵, R⁶ are each independently selected from H and R^{X2}; at each occurrence, R^{X1} and R are each independently selected from halogen, —NO₂, —CN, C₁₋₆ alkyl, C₂₋₆ alkenyl, C₂₋₆ alkynyl, —OR⁷, —SR⁷, —NR⁷R⁸, —C(=O)OR⁷, —C(=O)NR⁷R⁸, —OC(=O)R⁷, —NC(=O)R⁷R⁸, —C(=O)R⁷, —S(=O)₂OR⁷, —S(=O)₂R⁷, —S(=O)₂NR⁷R⁸, —OS(=O)₂R⁷, —NS(=O)₂R⁷R⁸ and —S(=O)R⁷, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO₂, —CN, —OH, —O(C₁₋₆ alkyl), —O(C₃₋₆ cyclohydrocarbyl), —O(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C₁₋₆ alkyl), —S(C₃₋₆ cyclohydrocarbyl), —S(C₁₋₄ alkylene-C₃₋₆ cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C₁₋₄ alkylene)-(3- to 7-membered heterocyclyl), —NH₂, —NH(C₁₋₆ alkyl), —

N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, =O, —COOH and C.sub.1-6 alkyl; at each occurrence, R.sup.7, R.sup.8 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl); at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6-alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, —OR.sup.Y1, —SR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2, —C(=O)R.sup.Y1, —S(=O).sub.2OR.sup.Y1, —S(=O).sub.2R.sup.Y1, —S(=O).sub.2NR.sup.Y1R.sup.Y2 and —S(=O)R.sup.Y1, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, =O, =S, —OR.sup.Y3, —SR.sup.Y3, —NR.sup.Y3R.sup.Y4, —C(=O)R.sup.Y3, —C(=O)OR.sup.Y3 and —C(=O)NR.sup.Y3R.sup.Y4; at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-8 alkyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, —OH, —SH, —NH.sub.2, =O and —COOH; ring A and R.sup.1 are as defined in formula (III); provided that the compound of formula (VI) does not include the following structure:

(157) ##STR00041## wherein R.sup.4 is selected from —O(C.sub.1-6 alkyl) and —O(C.sub.1-4 alkylene-C.sub.6-10 aryl).

(158) In one embodiment, ring B is a saturated or unsaturated 6-membered heterocycle, wherein heterocycle contains 1 or 2 heteroatoms each independently selected from N and O. In another embodiment, ring B is dihydropyrimidine. In a preferred embodiment, ring B is selected from 1,6-dihydropyrimidine, 1,2-dihydropyrimidine and 1,4-dihydropyrimidine.

(159) In a more preferred embodiment, the A-B ring system is

(160) ##STR00042##

wherein Y is O or S. In a particular embodiment, the A-B ring system is

(161) ##STR00043##

(162) In yet another embodiment, ring B is 2H-pyran or 4H-pyran. In a preferred embodiment, A-B ring system is

(163) ##STR00044##

wherein Y is O or S. In a particular embodiment, A-B ring system is

(164) ##STR00045##

(165) In yet another embodiment, ring C is phenyl, wherein phenyl is optionally substituted with 1, 2, 3, 4 or 5 substituents each independently selected from R.sup.X1.

(166) In one embodiment, L.sup.1 is a bond. In another embodiment, L.sup.1 is a C.sub.1-C.sub.6 hydrocarbon chain. In a preferred embodiment, L.sup.1 is C.sub.1-C.sub.2 hydrocarbon chain.

(167) In yet another embodiment, R.sup.2 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, =O, =S, —OR.sup.a1, —SR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, —C(=O)NR.sup.a1R.sup.b1, —C(=O)R.sup.a1, —S(=O).sub.2OR.sup.a1, —S(=O).sub.2R.sup.a1, —S(=O).sub.2NR.sup.a1R.sup.b1 and —S(=O)R.sup.a1, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2, —C(=O)R.sup.a2, —S(=O).sub.2OR.sup.a2, —S(=O).sub.2R.sup.a2, —S(=O).sub.2NR.sup.a2R.sup.b2 and —S(=O)R.sup.a2. In another embodiment, R.sup.2 is selected from H, halogen, —NO.sub.2, —CN, =O, =S, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —O(C=O)(C.sub.1-6 alkyl), —O(C=O)(C.sub.3-6 cyclohydrocarbyl), —O(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(C=O)(3- to 7-membered heterocyclyl), —O(C=O)(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C.sub.1-6 alkyl), —S(C.sub.3-6 cyclohydrocarbyl), —S(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, —NH(C=O)(C.sub.1-6 alkyl), —NH(C=O)(C.sub.3-6 cyclohydrocarbyl), —NH(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —NH(C=O)(3- to 7-membered heterocyclyl), —C(=O)(C.sub.1-6 alkyl), —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)O(C.sub.3-6 cyclohydrocarbyl), —C(=O)O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —C(=O)O(3- to 7-membered heterocyclyl), —C(=O)O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —C(=O)NH.sub.2, —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —C(=O)NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —C(=O)N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —C(=O)NH(3- to 7-membered heterocyclyl), —C(=O)N(3- to 7-membered heterocyclyl).sub.2, —C(=O)NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), and —C(=O)N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, wherein the alkyl, alkylene, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl or are optionally substituted with one or more substituents selected from halogen, nitro, cyano, —OH, —SH, —NH.sub.2 and —COOH.

In yet another embodiment, R.sup.2 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, —OR.sup.a1, —SR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, —C(=O)NR.sup.a1R.sup.b1, —C(=O)R.sup.a1, —S(=O).sub.2OR.sup.a1, S(=O).sub.2R.sup.a1, —S(=O).sub.2NR.sup.a1R.sup.b1 and —S(=O)R.sup.a1, wherein the alkyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2 and —C(=O)R.sup.a2. In a preferred embodiment, R.sup.2 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl and —OH, wherein the alkyl is optionally substituted with one or more

substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl and —NR.sup.a2R.sup.b2. In a more preferred embodiment, R.sup.2 is selected from H, halogen, C.sub.1-6 alkyl and —OH, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C.sub.1-6 alkyl and —NR.sup.a2R.sup.b2. In a further embodiment, R.sup.2 is selected from H, C.sub.1-4 alkyl and —OH, wherein the alkyl is optionally substituted with one or more substituents selected from —NR.sup.a2R.sup.b2. In a particular embodiment, R.sup.2 is selected from H, C.sub.1-4 alkyl and —OH, wherein the alkyl is —CH.sub.2[CH(CH.sub.3).sub.2] and is optionally substituted with one or more substituents selected from —NR.sup.b2. In another embodiment, R.sup.2 is an alkyl substituted with —NR.sup.a1R.sup.b1. In a particular embodiment, R.sup.2 is (168) ##STR00046##

(169) In one embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OW, —SR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —C(=O)R.sup.7, —S(=O).sub.2OR.sup.7, —S(=O).sub.2R.sup.7, —S(=O).sub.2NR.sup.7R.sup.8, —OS(=O).sub.2R.sup.7, —NS(=O).sub.2R.sup.7R.sup.8 and —S(=O)R.sup.7, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C.sub.1-6 alkyl), —S(C.sub.3-6 cyclohydrocarbyl), —S(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, =O and —COOH. In a preferred embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.7, —SR.sup.7, —NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —OS(=O).sub.2R.sup.7 and —NS(=O).sub.2R.sup.7R.sup.8, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), and —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2. In a preferred embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, C.sub.1-6 alkyl, —OR.sup.7 and —NR.sup.7R.sup.8, more preferably each independently selected from halogen and —OR.sup.7, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2. In a more preferred embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from F, Cl, Br, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, and, optionally substituted C.sub.1-6 alkyl, wherein the optionally substituted C.sub.1-6 alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2. In a particular embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from F, Cl, Br, methyl, —OH and dimethylaminomethyl.

(170) At each occurrence, R^{sup.7}, R^{sup.8} are each independently selected from H, C_{sub.1-6} alkyl, C_{sub.2-6} alkenyl and C_{sub.2-6} alkynyl, wherein the alkyl, alkenyl, alkynyl are optionally substituted with one or more substituents selected from halogen, —NO_{sub.2}, —CN, C_{sub.1-6} alkyl, —OH, —O(C_{sub.1-6} alkyl), —NH_{sub.2}, —NH(C_{sub.1-6} alkyl), —N(C_{sub.1-6} alkyl)_{sub.2}, —COOH, —C(=O)O(C_{sub.1-6} alkyl), —C(=O)NH(C_{sub.1-6} alkyl), —C(=O)N(C_{sub.1-6} alkyl)_{sub.2}, —OC(=O)(C_{sub.1-6} alkyl), —NHC(=O)(C_{sub.1-6} alkyl) and —C(=O)(C_{sub.1-6} alkyl). In a preferred embodiment, at each occurrence, R^{sup.7}, R^{sup.8} are each independently selected from H and C_{sub.1-6} alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C_{sub.1-6} alkyl, —OH, —O(C_{sub.1-6} alkyl), —NH_{sub.2}, —NH(C_{sub.1-6} alkyl), —N(C_{sub.1-6} alkyl)_{sub.2}, —COOH, —C(=O)O(C_{sub.1-6} alkyl), —C(=O)NH(C_{sub.1-6} alkyl), —C(=O)N(C_{sub.1-6} alkyl)_{sub.2}, —OC(=O)(C_{sub.1-6} alkyl), —NHC(=O)(C_{sub.1-6} alkyl), and —C(=O)(C_{sub.1-6} alkyl). In a particular embodiment, at each occurrence, R^{sup.7}, R^{sup.8} are each independently selected from H and C_{sub.1-6} alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —N(C_{sub.1-6} alkyl)_{sub.2} and —COOH. In another particular embodiment, at each occurrence, R^{sup.7}, R^{sup.8} are each independently selected from H and C_{sub.1-6} alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen and —N(C_{sub.1-6} alkyl)_{sub.2}.

(171) In one embodiment, at each occurrence, R^{sup.a1}, R^{sup.b1}, R^{sup.c1}, R^{sup.a2}, R^{sup.b2}, R^{sup.c2} are each independently selected from H, C_{sub.1-6} alkyl, C_{sub.3-6} cyclohydrocarbyl, C_{sub.3-6} cyclohydrocarbyl-C_{sub.1-4} alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C_{sub.1-4} alkyl, C_{sub.6-10} aryl-C_{sub.1-4} alkyl, 5- to 10-membered heteroaryl-C_{sub.1-4} alkyl, —OR^{sup.Y1}, —SR^{sup.Y1}, —NR^{sup.Y1}R^{sup.Y2}, —C(=O)OR^{sup.Y1}, —C(=O)NR^{sup.Y1}R^{sup.Y2} and —C(=O)R^{sup.Y1}, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —OR^{sup.Y3}, —SR^{sup.Y3}, —NR^{sup.Y3}R^{sup.Y4}, —C(=O)R^{sup.Y3}, —C(=O)OR^{sup.Y3} and —C(=O)NR^{sup.Y3}R^{sup.Y4}. In a preferred embodiment, at each occurrence, R^{sup.a1}, R^{sup.b1}, R^{sup.c1}, R^{sup.a2}, R^{sup.b2}, R^{sup.c2} are each independently selected from H, C_{sub.1-6} alkyl, C_{sub.6-10} aryl-C_{sub.1-4} alkyl, 5- to 10-membered heteroaryl-C_{sub.1-4} alkyl, —OR^{sup.Y1}, —SR^{sup.Y1}, —NR^{sup.Y1}R^{sup.Y2}, —C(=O)OR^{sup.Y1}, —C(=O)NR^{sup.Y1}R^{sup.Y2} and —C(=O)R^{sup.Y1}, wherein the alkyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —OR^{sup.Y3} and —NR^{sup.Y3}R^{sup.Y4}. In another preferred embodiment, at each occurrence, R^{sup.a1}, R^{sup.b1}, R^{sup.c1}, R^{sup.a2}, R^{sup.b2}, R^{sup.c2} are each independently selected from H, C_{sub.1-6} alkyl, —OR^{sup.Y1}, —NR^{sup.Y1}R^{sup.Y2}, —C(=O)OR^{sup.Y1}, —C(=O)NR^{sup.Y1}R^{sup.Y2} and —C(=O)R^{sup.Y1}, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OR^{sup.Y3} and —NR^{sup.Y3}R^{sup.Y4}. In a more preferred embodiment, R^{sup.a1}, R^{sup.b1}, R^{sup.c1}, R^{sup.a2}, R^{sup.b2}, R^{sup.c2} are each independently selected from H, C_{sub.1-6} alkyl, —OR^{sup.Y1}, —NR^{sup.Y1}R^{sup.Y2} and —C(=O)R^{sup.Y1}, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OR^{sup.Y3} and —NR^{sup.Y3}R^{sup.Y4}. In a further embodiment, at each occurrence, at each occurrence, R^{sup.a1}, R^{sup.b1}, R^{sup.c1}, R^{sup.a2}, R^{sup.b2}, R^{sup.c2} are each independently selected from H, C_{sub.1-6} alkyl, —OR^{sup.Y1}, —NR^{sup.Y1}R^{sup.Y2} and —C(=O)R^{sup.Y1}, wherein the alkyl is optionally substituted with one or more substituents selected from halogen and —NR^{sup.Y3}R^{sup.Y4}. In a particular embodiment, at each occurrence, R^{sup.a1}, R^{sup.b1}, R^{sup.c1}, R^{sup.a2}, R^{sup.b2}, R^{sup.c2} are each independently selected from H, C_{sub.1-3} alkyl, —OH and p-methylbenzoyl; wherein the alkyl is optionally substituted with one or more substituents selected from halogen and —NH_{sub.2}.

(172) In another embodiment, at each occurrence, R^{sup.Y1}, R^{sup.Y2}, R^{sup.Y3}, R^{sup.Y4} are each independently selected from H, C_{sub.1-8} alkyl, C_{sub.3-10} cyclohydrocarbyl, C_{sub.3-10}

cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2, —COOH and C.sub.1-6 alkyl. In a preferred embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-6 alkyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, phenyl, phenyl-C.sub.1-4 alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2, —COOH and C.sub.1-6 alkyl. In a more preferred embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-6 alkyl, phenyl, phenyl-C.sub.1-4 alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2, —COOH and C.sub.1-6 alkyl. In a more preferred embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-6 alkyl, phenyl and phenyl-C.sub.1-4 alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen and C.sub.1-6 alkyl. In a particular embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H and p-methylphenyl.

(173) In another embodiment, R.sup.2 is selected from H, halogen, —NO.sub.2, —CN, =O, =S, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —O(C=O)(C.sub.1-6 alkyl), —O(C=O)(C.sub.3-6 cyclohydrocarbyl), —O(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(C=O)(3- to 7-membered heterocyclyl), —O(C=O)(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —SH, —S (C.sub.1-6 alkyl), —S(C.sub.3-6 cyclohydrocarbyl), —S(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, —NH(C=O)(C.sub.1-6 alkyl), —NH(C=O)(C.sub.3-6 cyclohydrocarbyl), —NH(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —NH(C=O)(3- to 7-membered heterocyclyl), —C(=O)(C.sub.1-6 alkyl), —C(=O)O(C.sub.1-6 alkyl), —C(=O)O(C.sub.3-6 cyclohydrocarbyl), —C(=O)O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —C(=O)O(3- to 7-membered heterocyclyl), —C(=O)O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —C(=O)NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —C(=O)N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —C(=O)NH(3- to 7-membered heterocyclyl), —C(=O)N(3- to 7-membered heterocyclyl).sub.2, —C(=O)NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), and —C(=O)N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, wherein the alkyl, alkylene, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, nitro, cyano, —OH, —SH and —NH.sub.2. In yet another embodiment, R.sup.2 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.3-6 cyclohydrocarbyl,

C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, —OR.sup.a1, —SR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, —C(=O)NR.sup.a1R.sup.b1, —C(=O)R.sup.a1, —S(=O).sub.2OR.sup.a1, —S(=O).sub.2R.sup.a1, —S(=O).sub.2NR.sup.a1R.sup.b1 and —S(=O)R.sup.a1, wherein the alkyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2 and —C(=O)R.sup.a2. In a preferred embodiment, R.sup.2 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl and —OH, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl and —NR.sup.a2R.sup.b2. In a more preferred embodiment, R.sup.2 is selected from H, halogen, C.sub.1-6 alkyl and —OH, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C.sub.1-6 alkyl and —NR.sup.a2R.sup.b2. In a further embodiment, R.sup.2 is selected from H, C.sub.1-4 alkyl and —OH, wherein the alkyl is optionally substituted with one or more substituents selected from —NR.sup.a2R.sup.b2. In a particular embodiment, R.sup.2 is selected from H, C.sub.1-4 alkyl and —OH, wherein the alkyl is —CH.sub.2[CH(CH.sub.3).sub.2] and is optionally substituted with one or more substituents selected from —NR.sup.a2R.sup.b2. In another embodiment, R.sup.2 is an alkyl substituted with —NR.sup.a1R.sup.B1. In a particular embodiment, R.sup.2 is

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(175) In one embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OW, —SR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —C(=O)R.sup.7, —S(=O).sub.2OR.sup.7, —S(=O).sub.2R.sup.7, —S(=O).sub.2NR.sup.7R.sup.8, —OS(=O).sub.2R.sup.7, —NS(=O).sub.2R.sup.7R.sup.8 and —S(=O)R.sup.7, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3 to 7-membered heterocyclyl), —SH, —S(C.sub.1-6 alkyl), —S(C.sub.3-6 cyclohydrocarbyl), —S(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, and =O. In a preferred embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.7, —SR.sup.7, —NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —OS(=O).sub.2R.sup.7 and —NS(=O).sub.2R.sup.7R.sup.8, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3 to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), and —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2. In a preferred embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, C.sub.1-6 alkyl, —OR.sup.7 and —NR.sup.7R.sup.8, more preferably each independently selected from halogen and —OR.sup.7, wherein the alkyl is optionally substituted with one or more

substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2. In a more preferred embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from F, Cl, Br, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, and optionally substituted C.sub.1-6 alkyl, wherein the optionally substituted C.sub.1-6 alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2. In a particular embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from F, Cl, Br, methyl, —OH and dimethylaminomethyl.

(176) At each occurrence, R.sup.7, R.sup.8 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl and C.sub.2-6 alkynyl, wherein the alkyl, alkenyl, alkynyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6-alkyl), —N(C.sub.1-6 alkyl).sub.2, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl). In a preferred embodiment, at each occurrence, R.sup.7, R.sup.8 are each independently selected from H and C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl). In a particular embodiment, at each occurrence, R.sup.7, R.sup.8 are each independently selected from H and C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen and —N(C.sub.1-6 alkyl).sub.2. In another particular embodiment, at each occurrence, R.sup.7, R.sup.8 are each independently selected from H and C.sub.1-6 alkyl, wherein alkyl is optionally substituted with one or more substituents selected from halogen and —N(C.sub.1-6 alkyl).sub.2.

(177) In one embodiment, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered hetero aryl-C.sub.1-4 alkyl, —OR.sup.Y1, —SR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2 and —C(=O)R.sup.Y1, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —OR.sup.Y3, —SR.sup.Y3, —NR.sup.Y3R.sup.Y4, —C(=O)R.sup.Y3, —C(=O)OR.sup.Y3 and —C(=O)NR.sup.Y3R.sup.Y4. In a preferred embodiment, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, —OR.sup.1, —SR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.1, —C(=O)NR.sup.Y1R.sup.Y2 and —C(=O)R.sup.Y1, wherein the alkyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —OR.sup.3 and —NR.sup.Y3R.sup.Y4. In another preferred embodiment, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, —OR.sup.1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2 and —C(=O)R.sup.Y1, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OR.sup.3 and —NR.sup.Y3R.sup.Y4. In a more preferred embodiment, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a3, R.sup.b2, R.sup.c2, are each independently selected from H, C.sub.1-6 alkyl, —OR.sup.Y1, —NR.sup.Y1R.sup.Y2 and —C(=O)R.sup.Y1, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OR.sup.3 and —NR.sup.Y3R.sup.Y4. In a further embodiment, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are

each independently selected from H, C.sub.1-6 alkyl, —OR.sup.1, —NR.sup.Y1R.sup.Y2 and —C(=O)R.sup.Y1, wherein the alkyl is optionally substituted with one or more substituents selected from halogen and —NR.sup.Y3R.sup.Y4. In a particular embodiment, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-3 alkyl, —OH and p-methylbenzoyl; wherein the alkyl is optionally substituted with one or more substituents selected from halogen and —NH.sub.2.

(178) In another embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-8 alkyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2 and C.sub.1-6 alkyl. In a preferred embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-6 alkyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, phenyl, phenyl-C.sub.1-4 alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2 and C.sub.1-6 alkyl. In a more preferred embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-6 alkyl, phenyl, phenyl-C.sub.1-4 alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2 and C.sub.1-6 alkyl. In a more preferred embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-6 alkyl, phenyl and phenyl-C.sub.1-4 alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen and C.sub.1-6 alkyl. In a particular embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H and p-methylphenyl.

(179) In one embodiment, R.sup.3, R.sup.6 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH and —O(C.sub.1-6 alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), and —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl). In a preferred embodiment, R.sup.3, R.sup.6 are each independently selected from H, halogen and —OH, more preferably is H or —OH.

(180) In one embodiment, R.sup.4, R.sup.5 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —NH(3- to 7-membered heterocyclyl), —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH.sub.2, —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl), —C(=O)(C.sub.1-6 alkyl) and =O. In another embodiment, R.sup.4, R.sup.5 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —

N(C.sub.1-6 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH.sub.2, —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl), —C(=O)(C.sub.1-6 alkyl) and =O. In another embodiment, R.sup.4, R.sup.5 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH and —O(C.sub.1-6 alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH.sub.2, —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl), —C(=O)(C.sub.1-6 alkyl) and =O. In another embodiment, R.sup.4, R.sup.5 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH.sub.2, —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl), —C(=O)(C.sub.1-6 alkyl) and =O.

(181) The structure described in the following formula (VII) falls within the scope of the structure described in formula (I), formula (III) and formula (VI). In another aspect of the present invention, provided is use of a compound of formula (VII), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder.

(182) ##STR00048## wherein X is O; Y, R.sup.3, R.sup.4, R.sup.5, R.sup.6 are as defined in formula (VI); ring C is

(183) ##STR00049##

wherein R.sup.14, R.sup.15, R.sup.16, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, —OR.sup.7, —SR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —C(=O)R.sup.7, —S(=O).sub.2OR.sup.7, —S(=O).sub.2R.sup.7, —S(=O).sub.2NR.sup.7R.sup.8, —OS(=O).sub.2R.sup.7, —NS(=O).sub.2R.sup.7R.sup.8 and —S(=O)R.sup.7; wherein R.sup.7, R.sup.8 are as defined in formula (VI); R.sup.2 is selected from H, halogen, C.sub.1-6 alkyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, wherein the alkyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2, —C(=O)R.sup.a2, —S(=O).sub.2OR.sup.a2, —S(=O).sub.2R.sup.a2, —S(=O).sub.2NR.sup.a2R.sup.b2 and —S(=O)R.sup.a2; wherein R.sup.a2, R.sup.b2, R.sup.c2 are as defined in formula (VI); provided that the compound of formula (VII) does not include the following structure:

(184) ##STR00050## wherein R.sup.4 is selected from —O(C.sub.1-6 alkyl) and —O(C.sub.1-4 alkylene-C.sub.6-10 aryl).

(185) In one embodiment, Y is O. When Y is O, formula (VII) is

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(187) In one embodiment, R.sup.2 is selected from H, halogen and C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2, —C(=O)R.sup.a2, —S(=O).sub.2OR.sup.a2, —S(=O).sub.2R.sup.a2,

—S(=O).sub.2NR.sup.a2R.sup.b2 and —S(=O)R.sup.a2. In one embodiment, R.sup.2 is selected from —OR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1 and —C(=O)NR.sup.a1R.sup.b1. In another embodiment, R.sup.2 is selected from H, halogen and C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —NH.sub.2 and —COOH.

(188) In one embodiment, at each occurrence, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl) and —C(=O)N(C.sub.1-6 alkyl).sub.2. In a preferred embodiment, at each occurrence, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H and C.sub.1-6-alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl) and —C(=O)N(C.sub.1-6 alkyl).sub.2.

(189) In one embodiment, R.sup.3 is selected from H, halogen, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2. In one embodiment, R.sup.3 is selected from H, halogen, C.sub.1-3 alkyl, —OH, —O(C.sub.1-3 alkyl), —NH.sub.2, —NH(C.sub.1-3 alkyl) and —N(C.sub.1-3 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-2 alkyl), —NH.sub.2, —NH(C.sub.1-2 alkyl) and —N(C.sub.1-2 alkyl).sub.2. In one embodiment, R.sup.3 is selected from H, halogen, methyl, —OH, —OCH.sub.3, —NH.sub.2, —NHCH.sub.3 and —N(CH.sub.3).sub.2, wherein the methyl is optionally substituted with substituents selected from halogen, —OH, —O(C.sub.1-2 alkyl), —NH.sub.2, —NH(C.sub.1-2 alkyl) and —N(C.sub.1-2 alkyl).sub.2. In one embodiment, R.sup.3 is selected from H, halogen, methyl, —OH, —OCH.sub.3, —NH.sub.2, —NHCH.sub.3 and —N(CH.sub.3).sub.2, wherein the methyl is optionally substituted with substituents selected from halogen, —OH, —OCH.sub.3, —NH.sub.2, —NHCH.sub.3 and —N(CH.sub.3).sub.2. In one embodiment, R.sup.3 is selected from H, halogen, methyl, —OH, —NH.sub.2 and —N(CH.sub.3).sub.2. In yet another embodiment, R.sup.3 is selected from H, halogen, C.sub.1-4 alkyl, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2 and —NH(C.sub.1-4 alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2 and —NH(C.sub.1-4 alkyl). In one embodiment, R.sup.3 is selected from H, halogen, methyl, —OH, —OCH.sub.3, —NH.sub.2 and —NHCH.sub.3, wherein the methyl is optionally substituted with substituents selected from halogen, —OH, —O(C.sub.1-2 alkyl), —NH.sub.2 and —NH(C.sub.1-2 alkyl). In one embodiment, R.sup.3 is selected from H, halogen, methyl, —OH, —OCH.sub.3, —NH.sub.2 and —NHCH.sub.3, wherein the methyl is optionally substituted with substituents selected from halogen, —OH, —OCH.sub.3, —NH.sub.2 and —NHCH.sub.3. In another embodiment, R.sup.3 is selected from H, halogen, C.sub.1-4 alkyl, —OH, —O(C.sub.1-4 alkyl) and —NH.sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-4 alkyl) and —NH.sub.2. In one embodiment, R.sup.3 is selected from H, halogen, methyl, —OH, —OCH.sub.3 and —NH.sub.2, wherein the methyl is optionally substituted with substituents selected from halogen, —OH, —O(C.sub.1-2 alkyl) and —NH.sub.2. In one embodiment, R.sup.3 is selected from H, halogen, methyl, —OH, —OCH.sub.3 and —NH.sub.2, wherein the methyl is optionally substituted with substituents selected from halogen, —OH, —OCH.sub.3 and —NH.sub.2. In one embodiment, R.sup.3 is selected from H, halogen, methyl, —OH, —OCH.sub.3 and —NH.sub.2, wherein the methyl is optionally substituted with substituents selected from halogen, —OH, —OCH.sub.3 and —NH.sub.2.

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substituted with substituent(s) selected from halogen, —OH, —OCH.sub.3 and —NHCH.sub.3. In one embodiment, R.sup.3 is selected from H, halogen, —OH, —NH.sub.2 and methyl, wherein the methyl is substituted by —N(CH.sub.3).sub.2. In yet another embodiment, R.sup.3 is selected from H, halogen, —OH, —O(C.sub.1-4 alkyl), and substituted C.sub.1-4 alkyl, wherein the substituted C.sub.1-4 alkyl is substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2 and —NH(C.sub.1-4 alkyl). In one embodiment, R.sup.3 is selected from H, halogen, —OH, —OCH.sub.3, and substituted methyl, wherein the substituted methyl is substituted with substituent(s) selected from halogen, —OH, —O(C.sub.1-2 alkyl) and —NH(C.sub.1-2 alkyl). In one embodiment, R.sup.3 is selected from H, halogen, —OH, —OCH.sub.3, and substituted methyl, wherein the substituted methyl is substituted with substituent(s) selected from halogen, —OH, —OCH.sub.3 and —NHCH.sub.3. In one embodiment, R.sup.3 is selected from H, halogen, —OH and methyl, wherein the methyl is substituted with —N(CH.sub.3).sub.2. In another embodiment, R.sup.3 is selected from H, halogen, —OH, —O(C.sub.1-4 alkyl), and substituted C.sub.1-4 alkyl, wherein the substituted C.sub.1-4 alkyl is substituted with one or more substituents selected from halogen, —NH.sub.2 and —NH(C.sub.1-4 alkyl). In one embodiment, R.sup.3 is selected from H, halogen, —OH and C.sub.1-2 alkyl, wherein the alkyl is substituted with one or more substituents selected from halogen and —NH(C.sub.1-2 alkyl). In one embodiment, R.sup.3 is selected from H, halogen, —OH and methyl, wherein the methyl is substituted with substituent(s) selected from halogen and —NH(C.sub.1-2 alkyl). In one embodiment, R.sup.3 is selected from H, halogen, —OH and methyl, wherein the methyl is substituted with substituent(s) selected from halogen and —NHCH.sub.3. In one embodiment, R.sup.3 is selected from H, F, Cl, methyl, —OH and —NH.sub.2. In another embodiment, R.sup.3 is selected from H, F, methyl, —OH and —NH.sub.2. In one embodiment, R.sup.3 is H. In one embodiment, R.sup.3 is —OH. In one embodiment, R.sup.3 is CH.sub.3.

(190) In one embodiment, R.sup.4 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2. In one embodiment, R.sup.4 is selected from H, halogen, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2, —NH(C.sub.1-4 alkyl), —N(C.sub.1-4 alkyl).sub.2, and substituted C.sub.1-4 alkyl, wherein the substituted C.sub.1-4 alkyl is substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2, —NH(C.sub.1-4 alkyl) and —N(C.sub.1-4 alkyl).sub.2. In one embodiment, R.sup.4 is selected from H, halogen, —OH, —O(C.sub.1-3 alkyl), —NH.sub.2, —NH(C.sub.1-3 alkyl), —N(C.sub.1-3 alkyl).sub.2, and substituted C.sub.1-3 alkyl, wherein the substituted C.sub.1-3 alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-2 alkyl), —NH.sub.2, —NH(C.sub.1-2 alkyl) and —N(C.sub.1-2 alkyl).sub.2. In one embodiment, R.sup.4 is selected from H, halogen, —OH, —NH.sub.2, —NH(C.sub.1-4 alkyl), substituted C.sub.1-4 alkyl, substituted —O(C.sub.1-4 alkyl), and substituted —N(C.sub.1-4 alkyl).sub.2, wherein the substituted C.sub.1-4 alkyl, substituted —O(C.sub.1-4 alkyl) and substituted —N(C.sub.1-4 alkyl).sub.2 are substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-2 alkyl), —NH.sub.2 and —NH(C.sub.1-4 alkyl). In one embodiment, R.sup.4 is selected from H, halogen, —OH, —O(C.sub.1-2 alkyl), —NH.sub.2, —NH(C.sub.1-2 alkyl), substituted C.sub.1-2 alkyl, and substituted —O(C.sub.1-2 alkyl), wherein the substituted C.sub.1-2 alkyl and substituted —O(C.sub.1-2 alkyl) are substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-2 alkyl), —NH.sub.2 and —NH(C.sub.1-2 alkyl). In one embodiment, R.sup.4 is selected from H, halogen, —OH, —OCH.sub.3, —NH.sub.2, —NHCH.sub.3, and substituted C.sub.1-2 alkyl, wherein the substituted C.sub.1-2 alkyl is substituted with one or more substituents selected from halogen, —OH, —OCH.sub.3, —NH.sub.2 and —NHCH.sub.3. In one embodiment, R.sup.4 is selected from H, halogen, —OH, —NH.sub.2, —NH(C.sub.1-4 alkyl), and substituted

C.sub.1-4 alkyl, wherein the substituted C.sub.1-4 alkyl is substituted with one or more substituents selected from halogen, —OH, —OCH.sub.3, —NH.sub.2 and —NH(C.sub.1-4 alkyl). In one embodiment, R.sup.4 is selected from H, halogen, —OH, —NH.sub.2, —NH(C.sub.1-2 alkyl), —N(C.sub.1-2 alkyl).sub.2, and substituted C.sub.1-2 alkyl, wherein the substituted C.sub.1-2 alkyl is substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-2 alkyl), —NH.sub.2 and —NH(C.sub.1-2 alkyl). In one embodiment, R.sup.4 is selected from H, halogen, —OH, —NH.sub.2, —NHCH.sub.3, and substituted C.sub.1-2 alkyl, wherein the substituted C.sub.1-2 alkyl is substituted with one or more substituents selected from halogen, —OH, —OCH.sub.3, —NH.sub.2 and —NHCH.sub.3. In one embodiment, R.sup.4 is selected from H, halogen, C.sub.1-4 alkyl, —OH and —O(C.sub.1-4 alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2, —NH(C.sub.1-4 alkyl) and —N(C.sub.1-4 alkyl).sub.2. In one embodiment, R.sup.4 is selected from H, halogen, C.sub.1-2 alkyl, —OH and —OCH.sub.3, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —OCH.sub.3, —NH.sub.2, —NHCH.sub.3 and —N(CH.sub.3).sub.2. In one embodiment, R.sup.4 is selected from H, halogen, C.sub.1-4 alkyl, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2, —NH(C.sub.1-4 alkyl) and —N(C.sub.1-4 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH and —O(C.sub.1-4 alkyl). In one embodiment, R.sup.4 is selected from H, halogen, C.sub.1-2 alkyl, —OH, —OCH.sub.3, —NH.sub.2, —NHCH.sub.3 and —N(CH.sub.3).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH and —OCH.sub.3. In one embodiment, R.sup.4 is selected from H, halogen, C.sub.1-4 alkyl, —OH and —O(C.sub.1-4 alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH and —O(C.sub.1-4 alkyl). In one embodiment, R.sup.4 is selected from H, halogen, C.sub.1-2 alkyl, —OH and —OCH.sub.3, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH and —OCH.sub.3. In one embodiment, R.sup.4 is selected from H, —OH and —OCH.sub.3. In one embodiment, R.sup.4 is H. In one embodiment, R.sup.4 is —OH.

(191) In one embodiment, R.sup.5 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl). In one embodiment, R.sup.5 is selected from H, halogen, C.sub.1-4 alkyl, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2, —NH(C.sub.1-4 alkyl) and —N(C.sub.1-4 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2, —NH(C.sub.1-4 alkyl) and —N(C.sub.1-4 alkyl).sub.2. In one embodiment, R.sup.5 is selected from H, halogen, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2, —NH(C.sub.1-4 alkyl), —N(C.sub.1-4 alkyl).sub.2, —C(=O)O(C.sub.1-4 alkyl), —C(=O)NH.sub.2, —C(=O)NH(C.sub.1-4 alkyl), —C(=O)N(C.sub.1-4 alkyl).sub.2, —OC(=O)(C.sub.1-4 alkyl), —NC(=O)(C.sub.1-4 alkyl).sub.2, —C(=O)(C.sub.1-4 alkyl), and substituted C.sub.1-4 alkyl, wherein the substituted C.sub.1-4 alkyl is substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2, —NH(C.sub.1-4 alkyl), —N(C.sub.1-4 alkyl).sub.2, —C(=O)O(C.sub.1-4 alkyl), —C(=O)NH.sub.2, —C(=O)NH(C.sub.1-4 alkyl), —C(=O)N(C.sub.1-4 alkyl).sub.2, —OC(=O)(C.sub.1-4 alkyl), —NC(=O)(C.sub.1-4 alkyl).sub.2 and —C(=O)(C.sub.1-4 alkyl). In one embodiment, R.sup.5 is selected from H, halogen, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2, —NH(C.sub.1-4 alkyl), —N(C.sub.1-4 alkyl).sub.2, and substituted C.sub.1-4 alkyl, wherein the substituted C.sub.1-4 alkyl is substituted with one or more

substituents selected from halogen, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2, —NH(C.sub.1-4 alkyl) and —N(C.sub.1-4 alkyl).sub.2. In one embodiment, R.sup.5 is selected from H, halogen, —O(C.sub.1-4 alkyl), —NH(C.sub.1-4 alkyl), —N(C.sub.1-4 alkyl).sub.2, and substituted C.sub.1-2 alkyl, wherein the substituted C.sub.1-2 alkyl is substituted with one or more substituents selected from halogen, —O(C.sub.1-4 alkyl), —NH(C.sub.1-4 alkyl) and —N(C.sub.1-4 alkyl).sub.2. In one embodiment, R.sup.5 is selected from H, halogen, —O(C.sub.1-4 alkyl), —NH(C.sub.1-4 alkyl), —N(C.sub.1-4 alkyl).sub.2, and substituted C.sub.1-2 alkyl, wherein the substituted C.sub.1-2 alkyl is substituted with one or more substituents selected from halogen, —OCH.sub.3, —NHCH.sub.3 and —N(CH.sub.3).sub.2. In one embodiment, R.sup.5 is selected from H, halogen, —O(C.sub.1-2 alkyl), —NH(C.sub.1-2 alkyl), —N(C.sub.1-2 alkyl).sub.2, and substituted C.sub.1-2 alkyl, wherein the substituted C.sub.1-2 alkyl is substituted with one or more substituents selected from halogen, —OCH.sub.3, —NHCH.sub.3 and —N(CH.sub.3).sub.2.

(192) In one embodiment, R.sup.6 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2. In another embodiment, R.sup.6 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH and —NH.sub.2. In one embodiment, R.sup.6 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-2 alkyl, —OH, —O(C.sub.1-2 alkyl), —NH.sub.2, —NH(C.sub.1-2 alkyl) and —N(C.sub.1-2 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH and —NH.sub.2. In yet another embodiment, R.sup.6 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-4 alkyl, —OH, —O(C.sub.1-4 alkyl), —NH.sub.2, —NH(C.sub.1-4 alkyl) and —N(C.sub.1-4 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more halogen. In another embodiment, R.sup.6 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-2 alkyl, —OH, —O(C.sub.1-2 alkyl), —NH.sub.2, —NH(C.sub.1-4 alkyl) and —N(C.sub.1-4 alkyl).sub.2. In one embodiment, R.sup.6 is selected from H, F, Cl, Br, C.sub.1-6 alkyl, —OH and —NH.sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH and —NH.sub.2. In another embodiment, R.sup.6 is selected from H, F, Cl, Br, methyl, —OH and —NH.sub.2. In a particular embodiment, R.sup.6 is H or —OH, especially H.

(193) In one embodiment, R.sup.3 is methyl, and R.sup.4 is selected from —OH, —NH.sub.2, —NH(C.sub.1-3 alkyl), —N(C.sub.1-3 alkyl).sub.2, and C.sub.1-3 alkyl substituted with —COOH, wherein the substituted C.sub.1-3 alkyl is substituted with one or more substituents selected from —OH, —NH.sub.2, —NH(C.sub.1-2 alkyl), —N(C.sub.1-2 alkyl).sub.2 and —COOH. In one embodiment, R.sup.3 is methyl, and R.sup.4 is selected from —NH.sub.2, —NH(C.sub.1-3 alkyl), —N(C.sub.1-3 alkyl).sub.2, and C.sub.1-3 alkyl substituted with —COOH, wherein the substituted C.sub.1-3 alkyl is substituted with one or more substituents selected from —OH, —NH.sub.2, —NH(C.sub.1-2 alkyl), —N(C.sub.1-2 alkyl).sub.2 and —COOH.

(194) In another embodiment, R.sup.3 is methyl and R.sup.5 is selected from —OH, —NH.sub.2, —NH(C.sub.1-3 alkyl), —N(C.sub.1-3 alkyl).sub.2, and C.sub.1-3 alkyl substituted with —COOH, wherein the substituted C.sub.1-3 alkyl is substituted with one or more substituents selected from —OH, —NH.sub.2, —NH(C.sub.1-2 alkyl), —N(C.sub.1-2 alkyl).sub.2 and —COOH. In another embodiment, R.sup.3 is methyl, and R.sup.5 is selected from —NH.sub.2, —NH(C.sub.1-3 alkyl), —N(C.sub.1-3 alkyl).sub.2, and C.sub.1-3 alkyl substituted with —COOH, wherein the substituted C.sub.1-3 alkyl is substituted with one or more substituents selected from —OH, —NH.sub.2, —NH(C.sub.1-2 alkyl), —N(C.sub.1-2 alkyl).sub.2 and —COOH.

(195) In one embodiment, R.sup.14, R.sup.15, R.sup.16, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-

6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —O(C=O)(C.sub.1-6 alkyl), —O(C=O)(C.sub.3-6 cyclohydrocarbyl), —O(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(C=O)(3- to 7-membered heterocyclyl), —O(C=O)(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C.sub.1-6 alkyl), —S(C.sub.3-6 cyclohydrocarbyl), —S(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, —NH(C=O)(C.sub.1-6 alkyl), —NH(C=O)(C.sub.3-6 cyclohydrocarbyl), —NH(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —NH(C=O)(3- to 7-membered heterocyclyl), —C(=O)(C.sub.1-6 alkyl), —C(=O)O(C.sub.1-6 alkyl), —C(=O)O(C.sub.3-6 cyclohydrocarbyl), —C(=O)O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —C(=O)O(3- to 7-membered heterocyclyl), —C(=O)O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —C(=O)NH.sub.2, —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —C(=O)NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —C(=O)N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —C(=O)NH(3- to 7-membered heterocyclyl), —C(=O)N(3- to 7-membered heterocyclyl).sub.2, —C(=O)NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), and —C(=O)N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, wherein the alkyl, alkylene, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from H, halogen, nitro, cyano, —OH, —SH, —NH.sub.2, =O and —COOH. In one embodiment, R.sup.14, R.sup.15, R.sup.16, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8 and —C(=O)R.sup.7. In one embodiment, R.sup.14, R.sup.15, R.sup.16, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —O(C=O)(C.sub.1-6 alkyl), —O(C=O)(C.sub.3-6 cyclohydrocarbyl), —O(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(C=O)(3- to 7-membered heterocyclyl), —O(C=O)(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, —NH(C=O)(C.sub.1-6 alkyl), —NH(C=O)(C.sub.3-6 cyclohydrocarbyl), —NH(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), and —NH(C=O)(3- to 7-membered heterocyclyl). In one embodiment, R.sup.14, R.sup.15, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —O(C=O)(C.sub.1-6 alkyl), —O(C=O)(C.sub.3-6 cyclohydrocarbyl), —O(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(C=O)(3- to 7-membered heterocyclyl), —O(C=O)(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered

heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, —NH(C=O)(C.sub.1-6 alkyl), —NH(C=O)(C.sub.3-6 cyclohydrocarbyl), —NH(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), and —NH(C=O)(3- to 7-membered heterocyclyl). In one embodiment, R.sup.14, R.sup.15, R.sup.16, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —O(C=O)(C.sub.1-6 alkyl), —O(C=O)(C.sub.3-6 cyclohydrocarbyl), —O(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(C=O)(3- to 7-membered heterocyclyl), and —O(C=O)(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl). In one embodiment, R.sup.14, R.sup.15, R.sup.16, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —O(C=O)(C.sub.1-6 alkyl), —O(C=O)(C.sub.3-6 cyclohydrocarbyl), —O(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(C=O)(3- to 7-membered heterocyclyl), and —O(C=O)(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl). In one embodiment, R.sup.14, R.sup.15, R.sup.16, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, —NH(C=O)(C.sub.1-6 alkyl), —NH(C=O)(C.sub.3-6 cyclohydrocarbyl), —NH(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), and —NH(C=O)(3- to 7-membered heterocyclyl). In one embodiment, R.sup.14, R.sup.15, R.sup.16, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, —NH(C=O)(C.sub.1-6 alkyl), —NH(C=O)(C.sub.3-6 cyclohydrocarbyl), —NH(C=O)(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl) and —NH(C=O)(3- to 7-membered heterocyclyl). In one embodiment, R.sup.14, R.sup.15, R.sup.16, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2 and —NH(C=O)(C.sub.1-6 alkyl). In one embodiment, R.sup.14, R.sup.15, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2 and —NH(C=O)(C.sub.1-6 alkyl). In one embodiment, R.sup.16 is H. In one embodiment, R.sup.16 is —OCH.sub.3.

(196) The structure described in the following formula (VIII) falls within the scope of the structure described in formula (I), formula (III) and formula (VII). In another aspect of the present invention, provided is a use of a compound of formula (VIII), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder

(197) ##STR00052## wherein ring C, Y, R.sup.2, R.sup.3, R.sup.4, R.sup.5, R.sup.6 are as defined in formula (VII).

(198) In one embodiment, ring C, Y, R², R³, R⁴, R⁵, R⁶ in formula (VIII) are as defined in formula (VII); provided that the compound of formula (VIII) does not include the following

structure:

(199) ##STR00053## wherein R^{sup.4} is selected from —O(C_{sub.1-6} alkyl) and —O(C_{sub.1-4} alkylene-C_{sub.6-10} aryl).

(200) In one embodiment, Y is O. When Y is O, formula (VIII) is

(201) ##STR00054##

(202) In a particular embodiment, the compound of formula (VIII) is selected from:

(203) ##STR00055##

(204) The structure described in the following formula (IX) falls within the scope of the structure described in formula (I), formula (III) and formula (VI). In another aspect of the present invention, there is provided use of a compound of formula (IX), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder,

(205) ##STR00056## wherein: at each occurrence, R^{sup.19} is each independently selected from halogen, —NO_{sub.2}, —CN, C_{sub.1-6} alkyl, —OR^{sup.7}, —SR^{sup.7}, —NR^{sup.7}R^{sup.8}, —C(=O)OR^{sup.7}, —C(=O)NR^{sup.7}R^{sup.8}, —OC(=O)R^{sup.7}, —NC(=O)R^{sup.7}R^{sup.8}, —C(=O)R^{sup.7}, —S(=O)_{sub.2}OR^{sup.7}, —S(=O)_{sub.2}R^{sup.7}, —S(=O)_{sub.2}NR^{sup.7}R^{sup.8}, —OS(=O)_{sub.2}R^{sup.7}, —NS(=O)_{sub.2}R^{sup.7}R^{sup.8} and —S(=O)R^{sup.7}, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO_{sub.2}, —CN, —OH, —O(C_{sub.1-6} alkyl), —NH_{sub.2}, —NH(C_{sub.1-6} alkyl), —N(C_{sub.1-6} alkyl)_{sub.2} and —COOH; m is 0, 1, 2, 3, 4 or 5; R^{sup.2}, R^{sup.3}, R^{sup.4}, R^{sup.5}, R^{sup.6} are as defined in formula (VI).

(206) In one embodiment, R^{sup.2} is selected from H, halogen, —NO_{sub.2}, —CN, C_{sub.1-6} alkyl, C_{sub.3-6} cyclohydrocarbyl, C_{sub.3-6} cyclohydrocarbyl-C_{sub.1-4} alkyl, 3- to 7-membered heterocyclyl, 3- to 7-, membered heterocyclyl-C_{sub.1-4} alkyl, —OR^{sup.a1}, —SR^{sup.a1}, —NR^{sup.a1}R^{sup.b1}, —C(=O)OR^{sup.a1}, —C(=O)NR^{sup.a1}R^{sup.b1}, —C(=O)R^{sup.a1}, —S(=O)_{sub.2}OR^{sup.a1}, —S(=O)_{sub.2}R^{sup.a1}, —S(=O)_{sub.2}NR^{sup.a1}R^{sup.b1} and —S(=O)R^{sup.a1}, wherein the alkyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, —NO_{sub.2}, —CN, C_{sub.1-6} alkyl, —OR^{sup.a2}, SR^{sup.a2}, —NR^{sup.a2}R^{sup.b2}, —C(=O)OR^{sup.a2}, —C(=O)NR^{sup.a2}R^{sup.b2} and —C(=O)R^{sup.a2}. In a preferred embodiment, R^{sup.2} is selected from H, halogen, —NO_{sub.2}, —CN, C_{sub.1-6} alkyl and —OH, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO_{sub.2}, —CN, C_{sub.1-6} alkyl and —NR^{sup.a2}R^{sup.b2}. In a more preferred embodiment, R^{sup.2} is selected from H, halogen, C_{sub.1-6} alkyl and —OH, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C_{sub.1-6} alkyl and —NR^{sup.a2}R^{sup.b2}. In a further embodiment, R^{sup.2} is selected from H, C_{sub.1-4} alkyl and —OH, wherein the alkyl is optionally substituted with one or more substituents selected from —NR^{sup.a2}R^{sup.b2}. In a particular embodiment, R^{sup.2} is selected from H, C_{sub.1-4} alkyl and —OH, wherein the alkyl is —CH_{sub.2}[CH(CH_{sub.3})_{sub.2}], and is optionally substituted with one or more substituents selected from —NR^{sup.a2}R^{sup.b2}. In another embodiment, R^{sup.2} is an alkyl substituted with —NR^{sup.a1}R^{sup.b1}. R^{sup.a1}, R^{sup.b1}, R^{sup.c1}, R^{sup.a2}, R^{sup.b2}, R^{sup.c2} are as defined in formula (VI). In a particular embodiment, R^{sup.2} is

(207) ##STR00057##

(208) In one embodiment, at each occurrence, R^{sup.19} is each independently selected from halogen, C_{sub.1-6} alkyl, —OH, —O(C_{sub.1-6} alkyl), —NH_{sub.2}, —NH(C_{sub.1-6} alkyl) and —N(C_{sub.1-6} alkyl)_{sub.2}, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C_{sub.1-6} alkyl), —NH_{sub.2}, —NH(C_{sub.1-6} alkyl) and —N(C_{sub.1-6} alkyl)_{sub.2}.

(209) In a particular embodiment, the compound of formula (IX) is

(210) ##STR00058##

(211) In another aspect of the present invention, provided is use of a compound of formula (I'), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder

(212) ##STR00059## wherein: ring A is a benzene ring; ring B is a saturated or unsaturated 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S; ring C is C.sub.6-10 aryl, which is optionally substituted with one or more substituents each independently selected from R.sup.X1; L.sup.1 is a bond, or is a C.sub.1-C.sub.6 hydrocarbon chain; R.sup.1 is =Y, wherein Y is O or S; R.sup.2 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.14 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, =O, =S, =NR.sup.a1, —OR.sup.a1, —SR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, —C(=O)NR.sup.a1R.sup.b1, C(=O)R.sup.a1, —S(=O).sub.2OR.sup.a1, —S(=O).sub.2R.sup.a1, —S(=O).sub.2NR.sup.a1R.sup.b1, —S(=O)R.sup.a1, —C(=S)OR.sup.a1, —C(=S)NR.sup.a1R.sup.b1, —C(=S)R.sup.a1, —P(=O)(OR.sup.a1)OR.sup.b1, —C(=NR.sup.a1)NR.sup.b1R.sup.c1, —OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, =O, =S, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2, —C(=O)R.sup.a2, —S(=O).sub.2OR.sup.a2, —S(=O).sub.2R.sup.a2, —S(=O).sub.2NR.sup.a2R.sup.b2, —S(=O)R.sup.a2 and —C(=NR.sup.a2)NR.sup.b2R.sup.c2; R.sup.3, R.sup.4, R.sup.5, R.sup.6 are each independently selected from H and R.sup.X2; at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, —OR.sup.7, —SR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —C(=O)R.sup.7, —S(=O).sub.2OR.sup.7, —S(=O).sub.2R.sup.7, —S(=O).sub.2NR.sup.7R.sup.8, —OS(=O).sub.2R.sup.7, —NS(=O).sub.2R.sup.7R.sup.8 and —S(=O)R.sup.7, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2, =O, —COOH and C.sub.1-6 alkyl; at each occurrence, R.sup.7, R.sup.8 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl; at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, —OR.sup.Y1, —SR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2, —C(=O)R.sup.Y1, —S(=O).sub.2OR.sup.Y1, —S(=O).sub.2R.sup.Y1, —S(=O).sub.2NR.sup.Y1R.sup.Y2 and —S(=O)R.sup.Y1, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from =O, =S, —OR.sup.Y3, —SR.sup.Y3, —NR.sup.Y3R.sup.Y4, —C(=O)R.sup.Y3, —C(=O)OR.sup.Y3 and —C(=O)NR.sup.Y3R.sup.Y4; at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-8 alkyl, —C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl,

5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, —OH, —SH, —NH.sub.2, =O and —COOH.

(213) In one embodiment, R.sup.1 is =O. In another embodiment, R.sup.1 is =S.

(214) In one embodiment, ring B is a saturated or unsaturated 6-membered heterocycle, wherein the heterocycle contains 1 or 2 heteroatoms each independently selected from N and O. In another embodiment, ring B is dihydropyrimidine. In a preferred embodiment, ring B is selected from 1,6-dihydropyrimidine, 1,2-dihydropyrimidine and 1,4-dihydropyrimidine.

(215) In a more preferred embodiment, A-B ring system is

(216) ##STR00060##

wherein Y is O or S. In a particularly preferred embodiment, A-B ring system is

(217) ##STR00061##

(218) In yet another embodiment, ring B is 2H-pyran or 4H-pyran. In a preferred embodiment, A-B ring system is

(219) ##STR00062##

wherein Y is O or S. In a particularly preferred embodiment, A-B ring system is

(220) ##STR00063##

(221) In yet another embodiment, ring C is phenyl, which is optionally substituted with 1, 2, 3, 4 or 5 substituents each independently selected from R.sup.X1.

(222) In one embodiment, L.sup.1 is a bond. In another embodiment, L.sup.1 is a C.sub.1-C.sub.6 hydrocarbon chain. In a preferred embodiment, L.sup.1 is a C.sub.1-C.sub.2 hydrocarbon chain.

(223) In yet another embodiment, R.sup.2 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, =O, =S, —OR.sup.a1, —SR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, —C(=O)NR.sup.a1R.sup.b1, —C(=O)R.sup.a1, —S(=O).sub.2OR.sup.a1, —S(=O).sub.2R.sup.a1, —S(=O).sub.2NR.sup.a1R.sup.b1 and —S(=O)R.sup.a1, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, =O, =S, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2, —C(=O)R.sup.a2, —S(=O).sub.2OR.sup.a2, —S(=O).sub.2R.sup.a2, —S(=O).sub.2NR.sup.a2R.sup.b2 and —S(=O)R.sup.a2. In a preferred embodiment, R.sup.2 is selected from H, halogen, —NO.sub.2, —CN, =O, =S, —COOH, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C.sub.1-6 alkyl), —S(C.sub.3-6 cyclohydrocarbyl), —S(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, —C(=O)(C.sub.1-6 alkyl), —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)O(C.sub.3-6 cyclohydrocarbyl), —C(=O)O(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —C(=O)O(3- to 7-membered heterocyclyl), —C(=O)O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —C(=O)NH.sub.2, —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —

C(=O)NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —C(=O)N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —C(=O)NH(3 to 7-membered heterocyclyl), —C(=O)N(3- to 7-membered heterocyclyl).sub.2, —C(=O)NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), and —C(=O)N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, wherein the alkyl, alkylene, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, nitro, cyano, —OH, —SH, —NH, =O and —COOH. In another preferred embodiment, R.sup.2 is selected from H, halogen, —NO.sub.2, —CN and C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl and —NR.sup.a2R.sup.b2. In a more preferred embodiment, R.sup.2 is selected from H, halogen and C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C.sub.1-6 alkyl and —NR.sup.a2R.sup.b2. In a further preferred embodiment, R.sup.2 is H or C.sub.1-4 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from —NR.sup.a2R.sup.b2. In a particularly preferred embodiment, R.sup.2 is H or C.sub.1-4 alkyl, wherein the alkyl is —CH[CH(CH.sub.3).sub.2]— and is optionally substituted with one or more substituents selected from —NR.sup.a2R.sup.b2.

(224) In one embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, —OR.sup.7, —SR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —C(=O)R.sup.7, —S(=O).sub.2OR.sup.7, —S(=O).sub.2R.sup.7, —S(=O).sub.2NR.sup.7R.sup.8, —OS(=O).sub.2R.sup.7, —NS(=O).sub.2R.sup.7R.sup.8 and —S(=O)R.sup.7, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2, =O, —COOH and C.sub.1-6 alkyl. In a preferred embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.7, —SR.sup.7, —NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —OS(=O).sub.2R.sup.7 and —NS(=O).sub.2R.sup.7R.sup.8. In a preferred embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, C.sub.1-6 alkyl, —OR.sup.7 and —NR.sup.7R.sup.8. In a more preferred embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, C.sub.1-6 alkyl, —OR.sup.7 and —NR.sup.7R.sup.8. In a further preferred embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen and —OR.sup.7. In a more preferred embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from C.sub.1, Br, —OH and —O(C.sub.1-6 alkyl). In a particularly preferred embodiment, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from Cl and —OH.

(225) At each occurrence, R.sup.7, R.sup.8 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl and C.sub.2-6 alkynyl. In a preferred embodiment, at each occurrence, R.sup.7, R.sup.8 are each independently selected from H and C.sub.1-6 alkyl. In a particularly preferred embodiment, at each occurrence, R.sup.7, R.sup.8 are each independently selected from H and C.sub.1-6 alkyl.

(226) In one embodiment, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, OR.sup.Y1, SR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2, —C(=O)R.sup.Y1, —S(=O).sub.2OR.sup.Y1, —S(=O).sub.2R.sup.Y1, —S(=O).sub.2NR.sup.Y1R.sup.Y2 and —S(=O)R.sup.Y1, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted

with one or more substituents selected from =O, =S, —OR.sup.Y3, —SR.sup.Y3, —NR.sup.Y3R.sup.Y4, —C(=O)R.sup.Y3, —C(=O)OR.sup.Y3 and —C(=O)NR.sup.Y3R.sup.Y4. In a preferred embodiment, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, —OR.sup.Y1, —SR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2 and —C(=O)R.sup.Y1, wherein the alkyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from —R.sup.Y3 and —NR.sup.Y3R.sup.Y4. In another preferred embodiment, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, —OR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2 and —C(=O)R.sup.Y1, wherein the alkyl is optionally substituted with one or more substituents selected from —R.sup.Y3 and —NR.sup.Y3R.sup.Y4. In a more preferred embodiment, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, —OR.sup.Y1, —NR.sup.Y1R.sup.Y2 and —C(=O)R.sup.Y1, wherein the alkyl is optionally substituted with one or more substituents selected from —NR.sup.Y3 and —NR.sup.Y3R.sup.Y4. In a further preferred embodiment, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, —OR.sup.Y1, —NR.sup.Y1R.sup.Y2 and —C(=O)R.sup.Y1, wherein the alkyl is optionally substituted with one or more substituents selected from —NR.sup.Y3R.sup.Y4. In a particularly preferred embodiment, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-3 alkyl, —OH and p-methylbenzoyl; wherein the alkyl is optionally substituted with one or more substituents selected from —NH.sub.2. (227) In another embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-8 alkyl, —C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2, =O, —COOH and C.sub.1-6 alkyl. In a preferred embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, phenyl, phenyl-C.sub.1-4 alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2, —COOH and C.sub.1-6 alkyl. In a more preferred embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-6 alkyl, phenyl, phenyl-C.sub.1-4 alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2, —COOH and C.sub.1-6 alkyl. In a more preferred embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-6 alkyl, phenyl and phenyl-C.sub.1-4 alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen and C.sub.1-6 alkyl. In a particularly preferred embodiment, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H and p-methylphenyl.

(228) In a particularly preferred embodiment, the compound of formula (I') is selected from:

(229) ##STR00064##

(230) In yet another aspect, the present invention provides use of pharmaceutical composition comprising a compound of any one of formula (I), formula (II), formula (III), formula (III'), formula (IV), formula (V), formula (VI), formula (VII), formula (VIII), formula (IX) and formula (I') in the preparation of a medicament for the prevention or treatment of a polyQ-related neurodegenerative disorder, wherein the pharmaceutical composition comprises a compound of any one of formula (I), formula (II), formula (III), formula (III'), formula (IV), formula (V), formula (VI), formula (VII), formula (VIII), formula (IX) and formula (I'), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, and at least one pharmaceutically acceptable carrier.

(231) In another aspect, the present invention provides a compound of any one of formula (I), formula (II), formula (III), formula (III'), formula (IV), formula (V), formula (VI), formula (VII), formula (VIII), formula (IX) and formula (I'), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, or a pharmaceutical composition thereof, which is used to prevent or treat a polyQ-related neurodegenerative disorder.

(232) In a further aspect, the present invention provides a method for preventing or treating a polyQ-related neurodegenerative disorder, which includes administering a compound of any one of formula (I), formula (II), formula (III), formula (III'), formula (IV), formula (V), formula (VI), formula (VII), formula (VIII), formula (IX) and formula (I'), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, or a pharmaceutical composition thereof to an individual in need.

(233) The compound of the present invention can be administered to patients orally or parenterally in the form of conventional preparations, for example, capsules, microcapsules, tablets, granules, powders, lozenges, pills, suppositories, injections, suspensions, syrups, patches, creams, lotions, ointments, gels, sprays, solutions, and emulsions. Suitable formulations can use conventional organic or inorganic additives and are prepared by commonly used methods. The organic or inorganic additives are, for example, excipients (for example, sucrose, starch, mannitol, sorbitol, lactose, glucose, cellulose, talc, calcium phosphate or calcium carbonate), binder (for example, cellulose, methylcellulose, hydroxymethylcellulose, polypropylpyrrolidone, polyvinylpyrrolidone, gelatin, gum acacia, polyethylene glycol, sucrose or starch), disintegrants (for example, starch, carboxymethylcellulose, hydroxypropyl starch, low-substituted hydroxypropylcellulose, sodium bicarbonate, calcium phosphate or calcium citrate), lubricants (for example, magnesium stearate, light anhydrous silicic acid, talc or sodium lauryl sulfate), flavoring agents (for example, citric acid, menthol, glycine or tangerine powder), preservatives (for example, sodium benzoate, sodium bisulfite, methyl paraben or propyl paraben), stabilizer (for example, citric acid, sodium citrate or acetic acid), suspending agents (for example, methylcellulose, polyvinylpyrrolidone or aluminum stearate), dispersant (for example, hydroxypropyl methylcellulose), diluent (for example, water) and base wax (for example, cocoa butter, white petrolatum or polyethylene glycol). For example, the effective amount of the compound in the pharmaceutical composition may be an amount that can achieve the desired effect.

(234) Dosage regimens can be adjusted to provide the desired optimal response. For example, when the medicine is administered in the form of injection, it can be administered as a single bolus injection, bolus injection and/or continuous infusion, etc. For example, several divided doses may be administered over time, or the dose may be proportionally reduced or increased as indicated by the urgent need of the treatment situation. It should be noted that the dose value may vary with the type and severity of the condition to be alleviated and may include single or multiple doses. Generally, the dose of treatment varies, depending on the considerations, such as the age, gender, and general health of the patient to be treated; the frequency of treatment and the nature of the desired effect; the degree of tissue damage; the duration of symptoms; and other variables that can be adjusted by individual physicians. It should be further understood that for any particular

individual, the specific dosing regimen should be adjusted over time according to the needs of the individual and the professional judgment of the person administering the composition or supervising the administration of the composition. The dosage and regimen of the pharmaceutical composition can be easily determined by a person of ordinary skill in the clinical field. For example, the composition or compound of the present invention may be administered in divided doses from 4 times a day to once every 3 days, and the dosage may be, for example, 0.01 to 1000 mg/time. The required dose can be administered one or more times to obtain the desired result. The pharmaceutical composition according to the present invention can also be provided in unit dosage forms.

(235) In one embodiment, the present invention provides a capsule containing the compound of the present invention without an additional carrier.

(236) The pharmaceutical composition of the present invention may be in the form of tablets, chewable tablets, capsules, solutions, parenteral solutions, lozenges, suppositories, and suspensions, etc. The composition can be formulated to contain a daily dose or an appropriate portion of the daily dose in a dosage unit, which can be a single tablet or capsule or a liquid of a suitable volume.

(237) In one embodiment, the solution is prepared from a water-soluble salt, such as hydrochloride. Generally, all compositions are prepared according to known methods in pharmaceutical chemistry. Capsules can be prepared by mixing the compound with a suitable carrier or diluent and filling an appropriate amount of the mixture into the capsule. Commonly used carriers and diluents include, but are not limited to, inert powdered substances, such as a variety of different starches, powdered cellulose, especially crystalline and microcrystalline cellulose, sugars like fructose, mannitol and sucrose, cereal flour, and similar edible powder.

(238) Tablets can be prepared by direct compression, wet granulation, or dry granulation. The preparation usually adds diluent, binder, lubricant and disintegrant and the compound. Typical diluents include, for example, various types of starch, lactose, mannitol, kaolin, calcium phosphate or sulfate, inorganic salts (such as sodium chloride) and powdered sugar. Powdered cellulose derivatives are also useful. Typical tablet binders are the following substances, such as starch, gelatin, and sugar (such as lactose, fructose, glucose, etc.). Natural and synthetic gums are also suitable, including gum acacia, alginate, methylcellulose, polyvinylpyrrolidone, etc. Polyethylene glycol, ethylcellulose and wax can also act as binders.

(239) Lubricants can be selected from such slippery solids such as talc, magnesium stearate and calcium stearate, stearic acid and hydrogenated vegetable oils. A tablet disintegrant swells when wet to break the tablet and release the compound. They include starch, clay, cellulose, algin and gum. More specifically, for example, corn and potato starch, methylcellulose, agar, bentonite, lignocellulose, powdered natural sponge, anion exchange resin, alginic acid, guar gum, citrus pomace, carboxymethylcellulose and sodium lauryl sulfate can be used. Tablets can be coated with sugar as a flavoring and sealing agent or coated with a film-forming protective agent to optimize the dissolution performance of the tablet. The composition can also be formulated into chewable tablets, for example, by adding some substances to the formulation, such as mannitol.

(240) When it is desired to be administered as a suppository, a typical base can be used. Cocoa butter is a traditional suppository base, which can be changed by adding wax to slightly increase its melting point. Especially water-miscible suppository bases including polyethylene glycols of various molecular weights are widely used.

(241) The effect of the compound can be delayed or prolonged by a suitable formulation. For example, slowly dissolving pellets of the compound can be prepared and added to tablets or capsules or used as a sustained release implantable device. The technology also includes preparing several pellets with different dissolution rates and filling the capsule with a mixture of pellets. Tablets or capsules can be coated with a film that resists dissolution for a predictable period. Even parenteral preparations can be prepared as a long-acting formulation by dissolving or suspending

the compound in an oily or emulsified vehicle that allows it to be slowly dispersed in the serum.

(242) In a preferred embodiment, the polyQ-related neurodegenerative disorder is selected from spinocerebellar ataxia type 1, spinocerebellar ataxia type 2, spinocerebellar ataxia type 3, spinocerebellar ataxia type 7, spinocerebellar ataxia type 12, spinocerebellar ataxia type 17, dentatorubral-pallidoluysian atrophy, Huntington's disease and spinal-bulbar muscular atrophy. In another preferred embodiment, the polyQ-related neurodegenerative disorder is selected from the group consisting of spinocerebellar ataxia type 1, spinocerebellar ataxia type 2, spinocerebellar ataxia type 3, spinocerebellar ataxia type 7, spinocerebellar ataxia type 12 and spinocerebellar ataxia type 17. In another preferred embodiment, the polyQ-related neurodegenerative disorders are selected from Huntington's disease, spinocerebellar ataxia, type 1 and spinocerebellar ataxia type 3. In a particular embodiment, the polyQ-related neurodegenerative disorder is Huntington's disease. In another particular embodiment, the polyQ-related neurodegenerative disorder is spinocerebellar ataxia type 1 or spinocerebellar ataxia type 3.

(243) The embodiments of the present invention can be listed as follows:

(244) [1] Use of a compound of formula (I''), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder.

(245) ##STR00065## wherein ring A is a benzene ring; ring B is a saturated or unsaturated 5-membered or 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S; ring C is selected from C.sub.6-10 aryl and 5- to 10-membered heteroaryl, wherein the aryl or heteroaryl are optionally substituted with one or more substituents each independently selected from R.sup.X1; L.sup.1 is a bond, or is a C.sub.1-C.sub.6 hydrocarbon chain; or, ring C is absent, and L.sup.1 is absent; R.sup.1 is =Y, wherein Y is O or S, or is OR.sup.7; at each occurrence, R.sup.2 is each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.14 alkyl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, =O, =S, =NR.sup.a1, —OR.sup.a1, —SR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, —C(=O)NR.sup.a1R.sup.b1, —C(=O)R.sup.a1, —S(=O).sub.2OR.sup.a1, —S(=O).sub.2R.sup.a1, —S(=O).sub.2NR.sup.a1R.sup.b1, —S(=O)R.sup.a1, —C(=S)OR.sup.a1, —C(=S)NR.sup.a1R.sup.b1, —C(=S)R.sup.a1, —P(=O)(OR.sup.a1)OR.sup.b1, —C(=NR.sup.a1)NR.sup.b1R.sup.c1, —OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, =O, =S, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2, —C(=O)R.sup.a2, —S(=O).sub.2OR.sup.a2, —S(=O).sub.2R.sup.a2, —S(=O).sub.2NR.sup.a2R.sup.b2, —S(=O)R.sup.a2 and —C(=NR.sup.a2)NR.sup.b2R.sup.c2; R.sup.3, R.sup.4, R.sup.5, R.sup.6 are each independently selected from H and R.sup.X2; at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, —OR.sup.7, —SR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —C(=O)R.sup.7, —S(=O).sub.2OR.sup.7, —S(=O).sub.2R.sup.7, —S(=O).sub.2NR.sup.7R.sup.8, —OS(=O).sub.2R.sup.7, —NS(=O).sub.2R.sup.7R.sup.8 and —S(=O)R.sup.7, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), —O(C.sub.1-4 alkylene)-C.sub.3-6 cyclohydrocarbyl, —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C.sub.1-6 alkyl), —S(C.sub.3-6 cyclohydrocarbyl), —S(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —

NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —N(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, =O, —COOH and C.sub.1-6 alkyl; at each occurrence, R.sup.7, R.sup.8 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, and C.sub.6-10 aryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl or aryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl); at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6-alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, —OR.sup.Y1, —SR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2, —C(=O)R.sup.Y1, —S(=O).sub.2OR.sup.Y1, —S(=O).sub.2R.sup.Y1, —S(=O).sub.2NR.sup.Y1R.sup.Y2 and —S(=O)R.sup.Y1, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, =O, =S, —OR.sup.Y3, —SR.sup.Y3, —NR.sup.Y3R.sup.Y4, —C(=O)R.sup.Y3, —C(=O)OR.sup.Y3 and —C(=O)NR.sup.Y3R.sup.Y4; at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-8 alkyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.14 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, —OH, —SH, —NH.sub.2, =O and —COOH; n is 1 or 2; provided that the compound of formula (I'') does not include the following structure:

(246) ##STR00066## wherein R.sup.3 is selected from —O(C.sub.1-6 alkyl) and —O(C.sub.1-4 alkylene-C.sub.6-10 aryl); R.sup.10 is H; and the compound of formula (I'') does not include the following structure:

(247) ##STR00067## wherein R.sup.4 is —O(C.sub.1-6 alkyl); R.sup.5 is halogen; R.sup.10 is CF.sub.3; and the compound of formula (I'') does not include the following structure:

(248) ##STR00068## wherein R.sup.4 is selected from —O(C.sub.1-6 alkyl) and —O(C.sub.1-4 alkylene-C.sub.6-10 aryl); ring C is C.sub.6-10 aryl, which is optionally substituted with one or more substituents each independently selected from R.sup.X1; and the compound of formula (I'') does not include the following structure:

(249) ##STR00069## wherein R.sup.4 is selected from —OH and —O(C.sub.1-6 alkyl); R.sup.5 is halogen; ring C is 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl which is optionally substituted with one or more substituents each independently selected from R.sup.X1.

(250) [2] Use of [1] above, wherein the compound of formula (I'') has the following structure of formula (II''):

(251) ##STR00070## wherein Y is O or S; ring C is 5- to 7-membered heteroaryl, wherein the 5- to 7-membered heteroaryl is optionally substituted with one or more substituents each independently selected from R.sup.X1; R.sup.2 is selected from H and C.sub.1-8 alkyl; L.sup.1 is a bond, or is a

C.sub.1-C.sub.6 hydrocarbon chain; R.sup.3, R.sup.4, R.sup.5, R.sup.6 are each independently selected from H and R.sup.X2; wherein R.sup.X1, R.sup.X2 are as defined in [1] above.

(252) [3] Use of [1] above, wherein the compound of formula (I'') has the following structure of formula (IV''):

(253) ##STR00071## wherein Y is O or S; X is O; R.sup.9 is selected from H, halogen, C.sub.1-6 alkyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, —OR.sup.a1, —SR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, —C(=O)NR.sup.a1R.sup.b1, —C(=O)R.sup.a1, —S(=O).sub.2OR.sup.a1, —S(=O).sub.2R.sup.a1, —S(=O).sub.2NR.sup.a1R.sup.b1 and —S(=O)R.sup.a1, wherein the alkyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2, —C(=O)R.sup.a2, —S(=O).sub.2OR.sup.a2, —S(=O).sub.2R.sup.a2, —S(=O).sub.2NR.sup.a2R.sup.b2 and —S(=O)R.sup.a2; wherein R.sup.a1, R.sup.b1, R.sup.a2, R.sup.b2 are as defined in [1] above; R.sup.10 is selected from H, halogen, C.sub.1-6 alkyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl; R.sup.3 is selected from H, halogen, C.sub.1-6 alkyl, —OH, —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2; R.sup.4 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.7, —SR.sup.7 and —NR.sup.7R.sup.8; at each occurrence, R.sup.7, R.sup.8 are each independently selected from H and C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl); wherein R.sup.7, R.sup.8 are as defined in [1] above; R.sup.5 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl), wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl); R.sup.6 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —O(benzyl), —SH, —S(C.sub.1-6 alkyl), —S(benzyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2 and —NH(benzyl), wherein the alkyl or benzyl is optionally substituted with one or more substituents selected from halogen, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl) and —N(C.sub.1-6 alkyl).sub.2; provided that when R.sup.4 is —O(C.sub.1-6 alkyl) and R.sup.5 is halogen, R.sup.10 is not CF.sub.3.

(254) [4] Use of [1] above, wherein the compound of formula (I'') has the following structure of formula (VI''):

(255) ##STR00072## wherein: ring B is a saturated or unsaturated 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S; ring C is C.sub.6-10 aryl, which is substituted with one or more substituents each independently selected from R.sup.X1; L.sup.1 is a bond, or is a C.sub.1-C.sub.6 hydrocarbon chain; R.sup.2 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl-C.sub.1-

4 alkyl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, =O, =S, =NR.sup.a1, —OR.sup.a1, —SR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, —C(=O)NR.sup.a1R.sup.b1, —C(=O)R.sup.a1, —S(=O).sub.2OR.sup.a1, —S(=O).sub.2R.sup.a1, —S(=O).sub.2NR.sup.a1R.sup.b1, —S(=O)R.sup.a1, —C(=S)OR.sup.a1, —C(=S)NR.sup.a1R.sup.b1, —C(=S)R.sup.a1, —P(=O)(OR.sup.a1)OR.sup.b1, —C(=NR.sup.a1)NR.sup.b1R.sup.c1, —OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, =O, =S, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2, —C(=O)R.sup.a2, —S(=O).sub.2OR.sup.a2, —S(=O).sub.2R.sup.a2, —S(=O).sub.2NR.sup.a2R.sup.b2, —S(=O)R.sup.a2 and —C(=NR.sup.a2)NR.sup.b2R.sup.c2; R.sup.3, R.sup.4, R.sup.5, R.sup.6 are each independently selected from H and R.sup.X2; at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —C(=O)R.sup.7, —S(=O).sub.2OR.sup.7, —S(=O).sub.2R.sup.7, —S(=O).sub.2NR.sup.7R.sup.8, —OS(=O).sub.2R.sup.7, —NS(=O).sub.2R.sup.7R.sup.8 and —S(=O)R.sup.7, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —O(C.sub.3-6 cyclohydrocarbyl), alkylene-C.sub.3-6 cyclohydrocarbyl, —O(3- to 7-membered heterocyclyl), —O(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —SH, —S(C.sub.1-6 alkyl), —S(C.sub.3-6 cyclohydrocarbyl), —S(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), —S(3- to 7-membered heterocyclyl), —S(C.sub.1-4 alkylene)-(3- to 7-membered heterocyclyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —NH(C.sub.3-6 cyclohydrocarbyl), —N(C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(C.sub.1-4 alkylene-C.sub.3-6 cyclohydrocarbyl), alkylene-C.sub.3-6 cyclohydrocarbyl).sub.2, —NH(3- to 7-membered heterocyclyl), —N(3- to 7-membered heterocyclyl).sub.2, —NH(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl), —N(C.sub.1-4 alkylene-3- to 7-membered heterocyclyl).sub.2, =O, —COOH and C.sub.1-6 alkyl; at each occurrence, R.sup.7, R.sup.8 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl or heterocyclyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2, —COOH, —C(=O)O(C.sub.1-6 alkyl), —C(=O)NH(C.sub.1-6 alkyl), —C(=O)N(C.sub.1-6 alkyl).sub.2, —OC(=O)(C.sub.1-6 alkyl), —NHC(=O)(C.sub.1-6 alkyl) and —C(=O)(C.sub.1-6 alkyl); at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6-alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, —OR.sup.Y1, —SR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2, —C(=O)R.sup.Y1, —S(=O).sub.2OR.sup.Y1, —S(=O).sub.2R.sup.Y1, —S(=O).sub.2NR.sup.Y1R.sup.Y2 and —S(=O)R.sup.Y1, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, =O, =S, —OR.sup.Y3, —SR.sup.Y3, —NR.sup.Y3R.sup.Y4, —C(=O)R.sup.Y3, —C(=O)OR.sup.Y3 and —C(=O)NR.sup.Y3R.sup.Y4; at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-8 alkyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl,

C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, —OH, —SH, —NH.sub.2, =O and —COOH; ring A, R.sup.1 is as defined in [1] above; provided that the compound of formula (VI'') does not include the following structure:

(256) ##STR00073## wherein R.sup.4 is selected from —O(C.sub.1-6 alkyl) and —O(C.sub.1-4 alkylene-C.sub.6-10 aryl).

(257) [5] Use of [1] above, wherein ring B is dihydropyrimidine.

(258) [6] Use of [1] above, wherein ring B is 2H-pyran or 4H-pyran; preferably is 2H-pyran.

(259) [7] Use of any of [1]-[6] above, wherein R.sup.1 is =O.

(260) [8] Use of [5] above, wherein A-B ring system is

(261) ##STR00074##

(262) [9] Use of [6] above, wherein A-B ring system is

(263) ##STR00075##

(264) [10] Use of any of [4]-[9] above, wherein L.sup.1 is a bond, or is a C.sub.1-C.sub.2 hydrocarbon chain.

(265) [11] Use of any of [4]-[10] above, wherein ring C is phenyl, wherein the phenyl is optionally substituted with 1, 2, 3, 4 or 5 substituents each independently selected from R.sup.X1; preferably ring C is

(266) ##STR00076##

wherein R.sup.14, R.sup.15, R.sup.16, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, —OR.sup.7, —SR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —C(=O)R.sup.7, —S(=O).sub.2OR.sup.7, —S(=O).sub.2R.sup.7, —S(=O).sub.2NR.sup.7R.sup.8, —OS(=O).sub.2R.sup.7, —NS(=O).sub.2R.sup.7R.sup.8 and —S(=O)R.sup.7; preferably, R.sup.14, R.sup.15, R.sup.16, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8 and —C(=O)R.sup.7; more preferably, R.sup.14, R.sup.15, R.sup.17, R.sup.18 are each independently selected from H, halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OH, —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2 and —NH(C=O)(C.sub.1-6 alkyl); R.sup.16 is H or —OCH.sub.3.

(267) [12] Use of [4] above, wherein the compound of formula (VI'') has the structure of the following formula (IX''):

(268) ##STR00077## wherein: at each occurrence, R.sup.19 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, —OR.sup.7, —SR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —C(=O)R.sup.7, —S(=O).sub.2OR.sup.7, —S(=O).sub.2R.sup.7, —S(=O).sub.2NR.sup.7R.sup.8, —OS(=O).sub.2R.sup.7, —NS(=O).sub.2R.sup.7R.sup.8 and —S(=O)R.sup.7, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —O(C.sub.1-6 alkyl), —NH.sub.2, —NH(C.sub.1-6 alkyl), —N(C.sub.1-6 alkyl).sub.2 and —COOH; m is 0, 1, 2, 3, 4 or 5; R.sup.2, R.sup.3, R.sup.4, R.sup.5, R.sup.6 are as defined in [4] above.

(269) [13] Use of any of [1]-[12] above, wherein the compounds are selected from

(270) ##STR00078##

(271) [14] Use of a pharmaceutical composition in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorders; wherein the pharmaceutical composition comprises a compound of formula (I''), formula (II''), formula (III''), formula (III''), formula (IV''), formula (V''), formula (VI''), formula (VII''), formula (VIII''), formula (IX''), formula (X'') or

formula (XI'') in any one of [1]-[13] above, or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, and at least one pharmaceutically acceptable carrier.

(272) [15] Use of any of [1]-[14] above, wherein the polyQ-related neurodegenerative disorder is selected from the group consisting of spinocerebellar ataxia type 1, spinocerebellar ataxia type 2, spinocerebellar ataxia type 3, spinocerebellar ataxia type 7, spinocerebellar ataxia type 12, spinocerebellar ataxia type 17, dentatorubral-pallidoluysian atrophy, Huntington's disease, and spinal-bulbar muscular atrophy, especially Huntington's disease and spinocerebellar ataxia type 3.

(273) The embodiments of the present invention can also be listed as follows:

(274) <1> Use of a compound of formula (I'), or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorder

(275) ##STR00079## wherein: ring A is a benzene ring; ring B is a saturated or unsaturated 6-membered heterocycle, wherein the heterocycle contains 1, 2 or 3 heteroatoms each independently selected from N, O and S; ring C is C.sub.6-10 aryl, which is optionally substituted with one or more substituents each independently selected from R.sup.X1; L.sup.1 is a bond, or is a C.sub.1-C.sub.6 hydrocarbon chain; R.sup.1 is =Y, wherein Y is O or S; R.sup.2 is selected from H, halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.14 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, =O, =S, =NR.sup.a1, —OR.sup.a1, —SR.sup.a1, —NR.sup.a1R.sup.b1, —C(=O)OR.sup.a1, —C(=O)NR.sup.a1R.sup.b1, C(=O)R.sup.a1, —S(=O).sub.2OR.sup.a1, —S(=O).sub.2R.sup.a1, —S(=O).sub.2NR.sup.a1R.sup.b1, —S(=O)R.sup.a1, —C(=S)OR.sup.a1, —C(=S)NR.sup.a1R.sup.b1, —C(=S)R.sup.a1, —P(=O)(OR.sup.a1)OR.sup.b1, —C(=NR.sup.a1)NR.sup.b1R.sup.c1, OCN, —SCN, —N=C=O and —NCS, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, =O, =S, —OR.sup.a2, —SR.sup.a2, —NR.sup.a2R.sup.b2, —C(=O)OR.sup.a2, —C(=O)NR.sup.a2R.sup.b2, —C(=O)R.sup.a2, —S(=O).sub.2OR.sup.a2, —S(=O).sub.2R.sup.a2, —S(=O).sub.2NR.sup.a2R.sup.b2, —S(=O)R.sup.a2 and —C(=NR.sup.a2)NR.sup.b2R.sup.c2; R.sup.3, R.sup.4, R.sup.5, R.sup.6 are each independently selected from H and R.sup.X2; at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, —OR.sup.7, —NR.sup.7R.sup.8, —C(=O)OR.sup.7, —C(=O)NR.sup.7R.sup.8, —OC(=O)R.sup.7, —NC(=O)R.sup.7R.sup.8, —C(=O)R.sup.7, —S(=O).sub.2OR.sup.7, —S(=O).sub.2R.sup.7, —S(=O).sub.2NR.sup.7R.sup.8, —OS(=O).sub.2R.sup.7, —NS(=O).sub.2R.sup.7R.sup.8 and —S(=O)R.sup.7, wherein the alkyl, alkenyl or alkynyl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2, =O, —COOH and C.sub.1-6 alkyl; at each occurrence, R.sup.7, R.sup.8 are each independently selected from H, C.sub.1-6 alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, and 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl; at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6-alkyl, C.sub.2-6 alkenyl, C.sub.2-6 alkynyl, C.sub.3-6 cyclohydrocarbyl, C.sub.3-6 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 7-membered heterocyclyl, 3- to 7-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, —OR.sup.Y1, —SR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2, —C(=O)R.sup.Y1, —S(=O).sub.2OR.sup.Y1, —S(=O).sub.2R.sup.Y1, —S(=O).sub.2NR.sup.Y1R.sup.Y2 and —

S(=O)R.sup.Y1, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from =O, =S, —OR.sup.Y3, —SR.sup.Y3, —NR.sup.Y3R.sup.Y4, —C(=O)R.sup.Y3, —C(=O)OR.sup.Y3 and —C(=O)NR.sup.Y3R.sup.Y4; at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-8 alkyl, —C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl, C.sub.3-10 cyclohydrocarbyl-C.sub.1-4 alkyl, 3- to 10-membered heterocyclyl, 3- to 10-membered heterocyclyl-C.sub.1-4 alkyl, C.sub.6-10 aryl, C.sub.6-10 aryl-C.sub.1-4 alkyl, 5- to 10-membered heteroaryl, and 5- to 10-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl, alkenyl, alkynyl, cyclohydrocarbyl, heterocyclyl, aryl or heteroaryl are optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-8 alkyl, C.sub.2-8 alkenyl, C.sub.2-8 alkynyl, —OH, —SH, —NH.sub.2, =O and —COOH.

(276) <2> Use of <1> above, wherein ring B is saturated or unsaturated 6-membered heterocycle; wherein the heterocycle contains one or two heteroatoms each independently selected from N and O.

(277) <3> Use of <1> or <2> above, wherein ring B is dihydropyrimidine.

(278) <4> Use of <1> or <2> above, wherein ring B is 2H-pyran or 4H-pyran; preferably is 2H-pyran.

(279) <5> Use of any one of <1> to <4> above, wherein R.sup.1 is =O.

(280) <6> Use of <3> above, wherein A-B ring system is

(281) ##STR00080##

(282) <7> Use of <4> above, wherein A-B ring system is

(283) ##STR00081##

(284) <8> Use of any one of <1> to <7> above, wherein L.sup.1 is a bond, or is a C.sub.1-C.sub.2 hydrocarbon chain.

(285) <9> Use of any one of <1> to <8> above, wherein R.sup.2 is selected from H, halogen, —NO.sub.2, —CN and C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, C.sub.1-6 alkyl and —NR.sup.a2R.sup.b2; preferably, R.sup.2 is selected from H, halogen and C.sub.1-6 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from halogen, C.sub.1-6 alkyl and —NR.sup.a2R.sup.b2, more preferably, R.sup.2 is H or C.sub.1-4 alkyl, wherein the alkyl is optionally substituted with one or more substituents selected from —NR.sup.a2R.sup.b2, especially preferably, R.sup.2 is H or C.sub.1-4 alkyl, wherein the alkyl is —CH[CH(CH.sub.3).sub.2]—, and is optionally substituted with one or more substituents selected from —NR.sup.a2R.sup.b2.

(286) <10> Use of <9> above, wherein R.sup.2 is H.

(287) <11> Use of any one of <1> to <10> above, wherein at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, C.sub.1-6 alkyl, —OR.sup.7 and —NR.sup.7R.sup.8; preferably, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen, C.sub.1-6 alkyl, —OW and —NR.sup.7R.sup.8; more preferably, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from halogen and —OW; further preferably, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from C.sub.1, Br, —OH and —O(C.sub.1-6 alkyl); especially preferably, at each occurrence, R.sup.X1 and R.sup.X2 are each independently selected from C.sub.1 and —OH; wherein at each occurrence, R.sup.7, R.sup.8 are each independently selected from H and C.sub.1-6 alkyl.

(288) <12> Use of any one of <1> to <11> above, wherein at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, —OR.sup.Y1, —NR.sup.Y1R.sup.Y2, —C(=O)OR.sup.Y1, —C(=O)NR.sup.Y1R.sup.Y2 and —C(=O)R.sup.Y1, wherein the alkyl is optionally substituted with one or more substituents selected from —OR.sup.Y3 and —NR.sup.Y3R.sup.Y4; preferably, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, —OR.sup.Y1, —NR.sup.Y1R.sup.Y2 and —C(=O)R.sup.Y1, wherein the alkyl is optionally

substituted with one or more substituents selected from —OR.sup.Y3 and —NR.sup.Y3R.sup.Y4; more preferably, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-6 alkyl, —OR.sup.Y1, —NR.sup.Y1R.sup.Y2 and —C(=O)R.sup.Y1, wherein the alkyl is optionally substituted with one or more substituents selected from —NR.sup.Y3R.sup.Y4; especially preferably, at each occurrence, R.sup.a1, R.sup.b1, R.sup.c1, R.sup.a2, R.sup.b2, R.sup.c2 are each independently selected from H, C.sub.1-3 alkyl, —OH and p-methylbenzoyl; wherein the alkyl is optionally substituted with one or more substituents selected from —NH.sub.2.

(289) <13> Use of any one of <1> to <12> above, wherein at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-6 alkyl, phenyl, phenyl-C.sub.1-4 alkyl, 5- to 6-membered heteroaryl, and 5- to 6-membered heteroaryl-C.sub.1-4 alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen, —NO.sub.2, —CN, —OH, —SH, —NH.sub.2, —COOH and C.sub.1-6 alkyl; preferably, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H, C.sub.1-6 alkyl, phenyl and phenyl-C.sub.1-4 alkyl, wherein the alkyl or phenyl is optionally substituted with one or more substituents selected from halogen and C.sub.1-6 alkyl; especially preferably, at each occurrence, R.sup.Y1, R.sup.Y2, R.sup.Y3, R.sup.Y4 are each independently selected from H and p-methylphenyl.

(290) <14> Use of any one of <1> to <13> above, wherein the compound of formula (I') is selected from:

(291) ##STR00082##

(292) <15> Use of pharmaceutical composition in the preparation of a medicament for preventing or treating a polyQ-related neurodegenerative disorders; the pharmaceutical composition comprises a compound of formula (I') in any one of <1> to <14> above, or a pharmaceutically acceptable salt, stereoisomer, solvate, polymorph, tautomer, isotopic compound, metabolite or prodrug thereof, and at least one pharmaceutically acceptable carrier.

(293) <16> Use of any one of <1> to <15> above, wherein the polyQ-related neurodegenerative disorder is selected from the group consisting of spinocerebellar ataxia type 1, spinocerebellar ataxia type 2, spinocerebellar ataxia type 3, spinocerebellar ataxia type 7, spinocerebellar ataxia type 12, spinocerebellar ataxia type 17, dentatorubral-pallidoluysian atrophy, Huntington's disease, and spinal-bulbar muscular atrophy, especially Huntington's disease and spinocerebellar ataxia type 3.

(294) Beneficial Effect

(295) The compound of the present invention has the effect of reducing the level of polyQ-GFP fusion protein with expanded polyQ in cells and does not reduce the level of polyQ-GFP fusion protein with shorter polyQ. Therefore, the compound of the present invention can reduce the level of polyQ protein with expanded polyQ in cells or tissues, thereby having a preventive or therapeutic effect on polyQ protein-related disorders.

(296) The inventors further found that the compound of the present invention have an unexpected regulatory effect on mHTT levels in cells and had good safety. Administration of the compound of the present invention can improve the survival rate and climbing performance of *Drosophila* in HD model. Administration of the compound of the present invention through intraventricular or intraperitoneal injection can allele-selectively reduce the level of mHTT in the cerebral cortices and striata of HD model mice without affecting the level of wtHTT which has important physiological functions, and improved behavioral defects in mice. In the HD patient induced stem cells (iPSC) derived neurons, the compound of the present invention reduces mHTT levels, rescues disease-related phenotypes, and delays neuronal apoptosis. Therefore, the compound of the present invention has good selectivity, therapeutic effects and safety, and is easy to pass through the BBB, potentially facilitating oral administration.

(297) In addition, the compound of the present invention can significantly reduce the level of

mutant ATXN3 protein in cells, so it can be used for the treatment or prevention of spinocerebellar ataxia type 3.

(298) Therefore, the compound of the present invention can be used to treat polyQ protein-related disorders and has the advantages of high selectivity and good safety, as well as a good prospect for development of an oral drug.

EXAMPLES

(299) Unless otherwise specified, the instruments and reagents used herein are all commercially available.

(300) In all statistical analyses in this article, * represents $p < 0.05$; ** represents $p < 0.01$; *** represents $p < 0.001$, **** represents $p < 0.0001$. For the comparison between the two groups, the statistical analysis method used is the two-tailed unpaired t test. For comparisons between three or more groups, two-tailed one-way ANOVA was applied when there is only one variable, and two-tailed two-way ANOVA was applied when there are two variables.

(301) Abbreviation

(302) TABLE-US-00001 ATXN ataxin HTT Huntingtin TBP TATA binding protein ATN atrophin fl full-length BBB blood-brain barrier AR androgen receptor BDNF brain derived neurotrophic factor Experimental Materials, Reagents and Method Steps

Compounds

(303) Compound 1: ispinesib, PubChem CID: 6851740, can be purchased from Selleck, catalog number S1452; Compound 2: PubChem CID 5398649, can be purchased from ChemDiv, catalog number D715-2435; Compound 3: Semaxanib, can be purchased from Selleck, CAS No. 194413-58-6; Compound 4: Su9516, can be purchased from Selleck, CAS No. 377090-84-1; Compound 5: can be purchased from Sinopharm Chemical Reagent Co. Ltd, CAS No. 779-30-6; Compound 6: can be purchased from TargetMol, CAS No. 842-01-3; Compound 7: can be purchased from ChemDiv.

Antibody

(304) The antibodies used for Western-blots, HTRF and/or immunofluorescence/immunohistochemistry are as follows: HTT antibody 2B7 (Weiss et al. Anal Biochem 2009, 395, 8-15), 4C9, ab1 (Sapp et al. J Biol Chem 2012, 287, 13487-13499) and MW1 (Ko et al. Brain research bulletin 2001, 56, 319-329) were prepared by the method of the prior art; the antibody S830 used for immunostaining to detect HTT aggregates was gifted from Dr. Gillian Bates; other antibodies were purchased from companies such as Millipore and Sigma.

(305) Preparation and Verification of Recombinant Human Full-Length HTT Protein

(306) (1) The human HTT gene (GenBank: NM_002111.8) with (CAG).sub.23 or (CAG).sub.73 was synthesized de novo by Genewiz Inc. The human HTT gene was cloned into the modified pCAG vector (from Addgene) with N-terminal protein A tag.

(307) (2) The plasmid was transfected into human embryonic kidney E293 cells for expression using polyethyleneimine (PEI, from Polysciences, 23966). The protein is purified with IgG monoclonal antibody-agarose (Smart-lifesciences, SA030010), digested with TEV protease to remove the protein A tag, and further purified with Mono Q and Superose 6 (5/150 GL) columns from GE healthcare. Validation is conducted using Coomassie blue staining and Western-blots.

(308) Cells Used for Tests

(309) Primary cultured cortical neurons: The brains of Hdh.sup.Q140/Q7 and Hdh.sup.Q7/Q7 newborn mice (P0) were dissected, digested, dissociated, and cultured.

(310) Some primary patient fibroblasts and wild-type cells are from HD patients (Q47, Q55) of the Mongolian Huntington's disease family and healthy controls (WT, Q19). SCA3 cell line was from patient (Q74). The HD Q68 fibroblast cell line was from Coriell Cell Repositories (Camden, N.J., USA). Immortalized fibroblasts and induced stem cells (iPSCs) are prepared from primary fibroblasts. HEK293T cells were from ATCC.

(311) Animals Used for Tests

(312) Huntington's Disease *Drosophila*

(313) Nervous system driver line elav-GAL4 (c155), HTT expressing lines UAS-fl-HTT-Q16 and UAS-fl-HTT-Q128 were from Bloomington *Drosophila* Stock Center (<http://flystocks.bio.indiana.edu/>) and kept in a 25° C. incubator.

(314) Crosses were set up between virgin female *Drosophila* carrying elav-GAL4 driver and the UAS-fl-HTT-Q16 or UAS-fl-HTT-Q128 male *Drosophila* to generate a transgenic *Drosophila* driven by elav-GAL4 to express human HTT full-length protein (Q16) or (Q128) in the nervous system.

(315) Huntington's Disease Mice

(316) Mice expressing wild-type HTT gene (Hdh.sup.Q7/Q7) were from the Marian Difiglia laboratory of Massachusetts General Hospital, Harvard University. The Q140 gene knock-in heterozygous mouse (Hdh.sup.Q140/Q7) was prepared according to the method of the prior art (Menalled et al., J Comp Neurol, 2003, 465: 11-26).

(317) Protein Analysis

(318) Homogeneous time-resolved fluorescence (HTRF) analysis: the cell or tissue lysate was diluted with the original lysis buffer PBS+1% (v/v) Triton X-100+1×cComplete™ protease inhibitor to lyse the sample, then HTRF assay buffer (50 mM NaH.sub.2PO.sub.4, 400 mM NaF, 0.1% BSA, 0.05% (v/v) Tween-20, 1% (v/v) Triton X-100, pH 7.4) was used to dilute the specified antibody pair for detection. In HTRF buffer, the donor antibody concentration was 0.023 ng/μL and the acceptor antibody concentration was 1.4 ng/μL.

(319) Determination of the amount of protein: the amount of protein was determined by the above method. Background correction was performed using blank sample. The protein concentration for all samples were determined to correct the loadings. Different protein concentrations or cell numbers per well were measured to ensure that the signals are within the linear range.

(320) Cell Analysis

(321) Immunofluorescence: cells were washed, fixed, permeabilized and blocked, then incubated with primary antibody overnight at 4° C., then washed three times with blocking buffer, and incubated with secondary antibody for 1 hour at room temperature, stained with DAPI, and mounted. Images were taken by Zeiss Axio Vert A1 confocal microscope. TUBB3 or co-localization was analyzed using ImageJ.

Example 1 Effects of the Compounds on the Levels of Proteins Containing Expanded Polyglutamine

(322) The inventors used HEK293T cells exogenously expressing polyQ-GFP fusion proteins (Q72-GFP or Q25-GFP) to detect compounds that can control the level of proteins containing expanded polyglutamine in the cells. The method was as follows:

(323) (1) PolyQ-GFP cDNA sequence (expressing Met-polyQ-GFP, wherein polyQ was Q72 or Q25) was synthesized de novo and subcloned into pcDNA vector. It was transfected into HEK293T cells (ATCC, CRL-3216) through forward transfection to obtain Q72-GFP expressing cells and Q25-GFP expressing cells, respectively. And the same method was used to obtain cells expressing Q53-GFP, cells expressing Q46-GFP, and cells expressing Q38-GFP.

(324) (2) The polyQ-GFP proteins expressed by (1) were purified separately to obtain Q72-GFP, Q53-GFP, Q46-GFP, Q38-GFP and Q25-GFP (SEQ ID NO: 1). Among them, Q72-GFP, Q53-GFP, Q46-GFP, Q38-GFP differed from the amino acid sequence of SEQ ID NO: 1 in the length of the expanded polyglutamine.

(325) (3) The Q72-GFP expressing cells and Q25-GFP expressing cells obtained in step (1) were used. The cells were treated with 100 nM compound 1, or 50 nM compound 2 for 2 days. Then the level of polyQ-GFP was measured by Incucyte fluorescence counting (FIG. 1). It was observed that compound 1 and compound 2 effectively reduced the level of the protein containing expanded polyglutamine (Q72-GFP) in HEK293T cells, but did not reduce the level of the protein containing the shorter polyglutamine (Q25-GFP). This result indicated that the compound of the present

invention can selectively reduce the level of abnormally amplified polyQ protein in cells.

Example 2 Interaction of the Compounds with Proteins Containing Expanded Polyglutamine

(326) 2.1 OI-RD Detection of the Affinity of Compounds to Proteins Containing Expanded Polyglutamine

(327) Compound chips were prepared using a contact microarray printer (SmartArrayer 136, CapitalBio Corporation) according to the prior art (Zhu et al., Sensors (Basel) 2016, 16(3), 378; Fei et al., J Biomed Opt 2010, 15(1), 016018). Each compound was printed in triplicate on the microarray. OI-RD was used to measure the affinity reaction parameters between the compounds and GFP or the polyQ-GFP protein obtained in step (2) of Example 1. These compounds did not bind to Q25-GFP or GFP, but binded to proteins with expanded polyglutamine. For example, compound 2 showed certain binding affinity to proteins with 38 or more glutamine repeats. The affinity reaction parameters are shown in Table 1.

(328) TABLE-US-00002 TABLE 1 The association rate constants, dissociation rate constants, and equilibrium dissociation constants of the affinity reactions of compound 2 with Q72-GFP, Q53-GFP, Q46-GFP, and Q38-GFP

	$K_{\text{sub.on}}$ (min ⁻¹ Math. nM) ⁻¹	$K_{\text{sub.off}}$ (min) ⁻¹	$K_{\text{sub.d}}$ (nM)
Q72-GFP	$6.20 \times 10^{\text{sup.}-5}$	$1.92 \times 10^{\text{sup.}-2}$	309.7
Q53-GFP	$1.08 \times 10^{\text{sup.}-4}$	$4.65 \times 10^{\text{sup.}-3}$	43.1
Q46-GFP	$7.65 \times 10^{\text{sup.}-6}$	$2.5^{\text{sup.}7} \times 10^{\text{sup.}-2}$	3359.5
Q38-GFP	$2.99 \times 10^{\text{sup.}-6}$	$1.5 \times 10^{\text{sup.}-2}$	5016.7

2.2 MST Detection of the Affinity of Compounds to Proteins Containing Expanded Polyglutamine

(329) Furthermore, the affinity of compound 2 with proteins containing expanded polyglutamine was detected with micro thermophoresis (MST) using Monolith NT.115 instrument from NanoTemper Technologies. The reaction buffer was 20 mM HEPES, pH 7.4, 150 mM NaCl, and protein concentration was 500 nM. Compound 2 showed no binding affinity with Q25-GFP, and its $K_{\text{sub.d}}$ with Q72-GFP is 2.80 μM .

(330) In summary, the results of OI-RD detection and MST detection indicated that the compound of the present invention selectively binded to proteins containing expanded polyglutamine.

Example 3 Effect of the Compounds with Full-Length mHTT

(331) 3.1 OI-RD Detection of Affinity Between Compounds and Full-Length HTT

(332) Compound chips were prepared using a contact microarray printer (SmartArrayer 136, CapitalBio Corporation) according to the prior art (Zhu et al., Sensors (Basel) 2016, 16(3), 378; Fei et al., J Biomed Opt 2010, 15(1), 016018). Each compound was printed in triplicate on the microarray. OI-RD was used to measure the affinity reaction parameters of the compounds with full-length mHTT (flHTT-Q73) and full-length HTT-Q23 (flHTT-Q23, SEQ ID NO: 2) (FIG. 2). The compound showed no binding affinity with flHTT-Q23, and the association rate constants, dissociation rate constants and equilibrium dissociation constants of the affinity reactions with flHTT-Q73 are shown in Table 2.

(333) TABLE-US-00003 TABLE 2 The association rate constants, dissociation rate constants and equilibrium dissociation constants of the affinity reactions of the compounds with flHTT-Q73

	$K_{\text{sub.on}}$	$K_{\text{sub.off}}$	$K_{\text{sub.d}}$ (min ⁻¹ Math. nM) ⁻¹	(min) ⁻¹	(nM)
compound 1	$6.43 \times 10^{\text{sup.}-5}$	$2.0^{\text{sup.}7} \times 10^{\text{sup.}-2}$	321.9		
compound 2	$4.81 \times 10^{\text{sup.}-4}$	$1.19 \times 10^{\text{sup.}-2}$	24.7		

3.2 MST Detection of the Affinity of the Compound with Full-Length HTT

(334) The affinity of the compounds with full-length HTT was verified with micro thermophoresis (MST) using Monolith NT.115 instrument from NanoTemper Technologies (FIG. 3). The reaction buffer was 20 mM HEPES, pH 7.4, 150 mM NaCl. Protein concentration was 500 nM. The compound of the present invention showed no binding affinity with flHTT-Q23. The $K_{\text{sub.d}}$ of compound 1 and flHTT-Q73 was 0.13 μM ; the $K_{\text{sub.d}}$ of compound 2 and flHTT-Q73 was 2.82 μM .

(335) In summary, the results of OI-RD measurement and MST assay both showed that the compound of the present invention selectively binded with flHTT-Q73. This conclusion was inherently consistent with the phenomenon that the compound selectively reduced protein levels

containing expanded polyglutamine in HEK293T cells, indicating that the compound's effect on the above-mentioned protein levels may be achieved by selectively binding with proteins containing expanded polyglutamine.

Example 4 Effects of the Compounds on Cortical Neurons in HD Model Mice

(336) 4.1 Effects on mHTT and wtHTT Levels

(337) (1) The primary cultured Hdh.sup.Q140/Q7 and Hdh.sup.Q7/Q7 mouse cortical neuron cells were treated with the compound for 2 days. Then the levels of mHTT and wtHTT were detected by Western-blot (antibody 2166) (FIG. 4 and FIG. 5). Compound 1 and compound 2 reduced the level of mHTT in cortical neurons of Q140 knock-in heterozygous mice (Hdh.sup.Q140/Q7), but hardly affected the level of wtHTT. Compound 1 did not reduce wtHTT levels in cortical neurons of Hdh.sup.Q7/Q7 mice.

(338) (2) mHTT was detected using anti-polyQ antibodies MW1 and 3B5H10, and observed for bands of proteins with smaller molecular weights. As a result, no increase in the N-terminal fragments of mHTT was observed (FIG. 6). The detected decrease in mHTT level was not due to an increase in site-specific cleavage.

(339) 4.2 Cytotoxicity Test

(340) CellTiter-glo (Promega, G7570) was used according to the protocol provided in the kit to determine the viability of Hdh.sup.Q140/Q7 mouse cortical neurons after treatment with the specified compound (FIG. 7).

(341) The compound of the present invention showed no cytotoxicity to Hdh.sup.Q140/Q7 mouse cortical neurons at the test concentration described in 4.1. The detected decrease in mHTT level was not due to neuronal cell loss.

(342) In summary, the compound of the present invention allele-selectively reduced the level of mHTT in cells, and had no cytotoxicity, and had good safety.

Example 5 Effects of the Compounds on mHTT and wtHTT Levels of Huntington's Disease Patient Fibroblasts

(343) 5.1 Effect on the Levels of mHTT and wtHTT in HD Patient Primary Fibroblasts

(344) Preliminary experiments were conducted. As a result, the compound exhibited the best mHTT reduction effect at a concentration of 100 nM.

(345) Experiment method: The fibroblasts (Q49, Q55, Q68) of HD patients were treated with 100 nM compound for 2 days. Then mHTT (antibody pair: 2B7/MW1) and total HTT (antibody pair: 2B7/2166) were detected by HTRF. The results are shown in FIG. 8. Reduced mHTT levels were observed in primary fibroblasts (Q49, Q55, Q68) of HD patients. No reduction in HTT levels was observed in wild-type primary fibroblasts.

(346) 5.2 Effects on the Level of mHTT in HD Patient Immortalized Fibroblasts

(347) A test method similar to that described in 5.1, HTRF (antibody pair: 2B7/MW1) was used to detect the effects of the compound of the present invention on the level of mHTT in HD patient immortalized fibroblasts (FIG. 9). Decreased mHTT levels were observed on HD patient immortalized fibroblasts. FIG. 10 shows that compound 2 reduced the level of mHTT in HD patient immortalized fibroblasts at different doses.

Example 6 Effects of the Compounds on mHTT Level and Neuronal Apoptosis of HD Patient iPSC-Derived Neurons

(348) 6.1 Effect on mHTT and wtHTT Levels

(349) Experimental Materials:

(350) A test method similar to that described in 5.1, HTRF (antibody pair: 2B7/MW1) was used to detect the effects of compound 1 and compound 2 on the level of mHTT in HD patient iPSC-derived neurons (Q47) (FIG. 11). Decreased mHTT levels were observed in HD patient iPSC-derived neurons.

(351) Using the same method, the following compound 3, compound 4, compound 5, compound 6, and compound 7 were observed to reduce the mHTT level in the HD patient iPSC-derived neurons;

wherein compound 3 reduced mHTT levels by approximately 12.2%, compound 4 reduced mHTT levels by approximately 24.9%, compound 5 reduced mHTT levels by approximately 26.8%, compound 6 reduced mHTT levels by approximately 24.2%, and compound 7 reduced mHTT levels by approximately 19.7%.

(352) The compound of the present invention rescued disease-related phenotypes in HD patient iPSC-derived neurons through autophagy.

(353) 6.2 Effects on Neuronal Apoptosis

(354) (1) Immunostaining

(355) HD patient iPSC-derived neurons (Q47) were treated using 100 nM compound 1 or 50 nM compound 2 for 1 day. Then the cells were stressed (by BDNF removal).

(356) Neuronal specific tubulin marker TUBB3 was stained with DAPI. The TUBB3 signal covered area was normalized to the cell nuclei counts, and the data was normalized to the wild type controls to analyze neuronal apoptosis (FIG. 12).

(357) (2) Caspase-3 Activity Detection

(358) After removing BDNF, loss of processes and shrinkage of neurons were observed in HD neurons.

(359) NucView 488 (Biotium, 30029) was used to detect active caspase-3. After removing BDNF, Incucyte (Essen Bioscience, IncuCyte FLR) was used to capture images every 3 hours in the incubator, which were analyzed with Incucyte 2011A software. Three batches were tested and the results were consistent (FIG. 13). Compound 1 and compound 2 significantly improved the loss of processes and shrinkage of HD neurons after removal of BDNF.

Example 7 Effects of the Compounds on Huntington's Disease *Drosophila*

(360) 7.1 Effects on mHTT Level

(361) Experimental Method:

(362) Q128 *Drosophila* and Q16 *Drosophila* were randomized into a negative control group and a positive drug group (compound 1, compound 2), with 75 *drosophila* in each group. The negative control group was given the corresponding solvent DMSO, and the positive drug groups were given the corresponding positive drug.

(363) *Drosophila* were kept in standard food at 25° C. The newly hatched *Drosophila* were transferred to vials with food containing the positive drug (10 µM in 400 µL DMSO) or DMSO for control. The corn food was changed every other day.

(364) After fed for 6 consecutive days, the *Drosophila* head protein was extracted on the 7th day. The mHTT level was measured by HTRF (antibody pair: 2B7/MW1), where each sample included head proteins from five *Drosophila* (FIG. 14). Compound 1 and compound 2 reduced mHTT levels in transgenic *Drosophila* expressing human HTT full-length protein (Q128).

(365) 7.2 Effects of Compound on Survival Rate

(366) 75 age-matched virgin *Drosophila* were put into empty plastic vials containing standard food, and the survival rate of each vial was recorded daily to measure the lifespan (FIG. 15). The survival rate of the Q128 *Drosophila* positive drug group was improved compared to the control group.

(367) 7.3 Effects on Climbing Performance

(368) 15 age-matched virgin *Drosophila* were put into empty vials and tapped down so that they were at the bottom of the vials. The percentage of *Drosophila* that had climbed past a 7-cm-high line after 15 seconds was recorded. The mean of five observations for each vial was plotted every day, and the data from multiple vials containing different batches of *Drosophila* was plotted and analyzed.

(369) The results are shown in FIG. 16. The number in brackets indicates the number of tested vials. The Q128 *Drosophila* positive drug group showed improved climbing performance compared to the control group.

(370) In the above experiments in 7.2 and 7.3, no compound was observed to have an effect on

Q16 Drosophilae.

Example 8 Effects of the Compounds on HD Model Mice

(371) Experimental animal: The mice were grouped and housed in individually vented cages with a 12-hour light/dark cycle, with a maximum of 5 adult mice per cage.

(372) 8.1 Effects of Intracerebroventricular Injection of Compounds on the Levels of mHTT and wtHTT in the Cortices of HD Mice

(373) Experimental animals: Hdh.sup.Q140/Q7 mice (3 months old), 4 in each group.

(374) Experimental method: one icv-injection was conducted per day using 2 μ L artificial cerebrospinal fluid (ACSF: 1 mM glucose, 119 mM NaCl, 2.5 mM KCl, 1.3 mM MgSO₄, 2.5 mM CaCl₂, 26.2 mM NaHCO₃, 1 mM NaH₂PO₄) containing a compound at a concentration of 25 μ M. 2 μ L ACSF containing the same amount of DMSO was used as a control.

(375) After 10 days of injection, the brain cortical neuron proteins of the mice were extracted, and the levels of mHTT and wtHTT were detected by Western-blots (antibody 2166). Each test includes three repetitions for each sample, and the mean values was calculated (FIG. 17). Compound 2 significantly reduced mHTT level in Hdh.sup.Q140/Q7 mouse cortices and showed mHTT selectivity relative to wtHTT.

(376) 8.2 Compound 2 Administered by Intraperitoneal Injection

(377) Experimental method: The compound or control DMSO was diluted with 0.9% NaCl intravenous infusion solution to 0.05 μ g/ μ L. One ip-injection (0.5 mg/kg) was conducted per day. After 14 days of injection, tissue extraction or behavioral experiment were conducted.

(378) In vivo compound detection in brain tissues of ip-injected mice: 2 hours after ip-injection of DMSO or compound in mice, the brain was dissected, weighed, and the compound in the brain tissue was extracted, analyzed with UPLC-MS (Acquity Ultra Performance Liquid Chromatography System, Acquity UPLC BEH C18 (1.7 μ m, 2.1 $\times$ 50 mm) column and Xevo TQ-S mass spectrometer, Waters Corporation, Milford, MA, USA) for LC-MS/MS analysis. The level of compound 2 reached the brain tissue was 9.48 ng/g. No mass spectrum signal of compound was observed for the control (DMSO) group. The compound of the present invention can be delivered to the brain of mice through the BBB smoothly, which potentially facilitates oral administration of the compound.

(379) 8.3 Effects of Intraperitoneal Injection of Compounds on the Levels of mHTT and wtHTT in the Cortices and Striata of HD Mice

(380) (1) 13 Hdh.sup.Q140/Q7 mice (5 months old) were randomized into 2 groups.

(381) Administer compound 2 as described in 8.2. One ip-injection was conducted per day. After 14 days of injection, the protein was extracted and the levels of mHTT and wtHTT in the cortices of HD mice were detected by Western-blots (FIG. 18).

(382) (2) A total of 13 mice Hdh.sup.Q140/Q7 mice (10 months old) were used, 6-7 mice in each group. According to the above method, the mouse brain striatum neuronal protein was extracted and the levels of mHTT and wtHTT were detected by Western-blots (FIG. 19).

(383) (3) mHTT aggregates in the cortices of Hdh.sup.Q140/Q7 mice were detected by dot-blot experiment (antibody: 4C9, bar plot showing the results of two repetitions) and HTRF (antibody pair: 4C9/4C.sub.9). Sampling was repeated for each mouse two to three times on average (FIG. 20). No increase in mHTT aggregates was observed. The decrease in mHTT levels in the compound 2 treated group was not due to changes in mHTT solubility.

(384) In summary, intraperitoneal injection of compound 2 reduced the levels of mHTT in the cortices and striata of Hdh.sup.Q140/Q7 mice, and showed mHTT selectivity relative to wtHTT. Therefore, compound 2 had the prospect of being developed to an oral drug.

(385) 8.4 Effects of Intraperitoneal Injection of Compound 2 on Behavioral Defects in HD Mice

(386) Experimental animals: 32 Hdh.sup.Q140/Q7 mice were randomized into 2 groups; 28 (Hdh.sup.Q7/Q7) mice were randomized into 2 groups.

(387) Experimental method: administer compound 2 according to the procedure described in 8.2.

One ip-injection was conducted per day, and behavioral experiments were conducted after 14 days of injection.

(388) All behavioral experiments were carried out during the light phase. Before starting the experiments, all mice were kept in the behavioral test room under dim red light for one hour.

(389) Rotarod test: The mice were pre-trained for 3 consecutive days (on the rotarod rotating at 4 rpm for 2 minutes). The result of each experiment was recorded as time on the rod (time on the rotating rod), until falling from the rod or until the end of the task. Each test include three repetitions with an inter-trial interval of 60 min in order to reduce stress and fatigue. The means of three trials were analyzed for each mouse (FIG. 21a).

(390) Balance beam test: a 2 cm thick meter stick with a total length of 100 cm was suspended on the platform on both sides. There was a bright light at the starting point and a dark box with food at the endpoint. The total time for each mouse to walk through the balance beam was recorded (FIG. 21b).

(391) The compound of the present invention improved the Huntington's disease related behavioral defects in HD model mice, and had no effect on wild-type mice.

Example 9 Effects of the Compounds and the ATXN3 with Abnormally Expanded polyQ

(392) 9.1 Preparation of Recombinant Human MBP-ATXN3

(393) (1) His8 tag and TEV protease cleavage site is added in pMal-C2x plasmid (from New England Biolabs) to prepare prokaryotic expression vector pMBP. The ATXN3 gene (GenBank: NM_001127696.2) is edited to control the length of the expanded polyglutamine, the gene is amplified and cloned into the prepared pMBP to obtain the pMBP-ATXN3-Q28 and pMBP-ATXN3-Q68 plasmids with polyQ repeated 28 and 68 times, respectively.

(394) (2) The expression plasmids pMBP-ATXN3-Q28 and pMBP-ATXN3-Q68 were introduced into *Escherichia coli* Rosetta(DE3)pLysS for expression. Preliminary purification was conducted with HisTrap HP column (GE Healthcare, 17524701). The product was concentrated by ultrafiltration and then further purified with Superose 6 Increase 10/300 GL size exclusion column (GE Healthcare).

(395) (3) As verified by SDS-PAGE, the purity of the prepared human MBP-ATXN3-Q28 and MBP-ATXN3-Q68 exceeds 98%.

(396) 9.2 MST Detection of the Affinity of the Compound to ATXN3 with Abnormally Expanded polyQ

(397) Following the method in 2.2, the affinity of compound 2 with MBP-ATXN3-Q28 and MBP-ATXN3-Q68 protein was verified by microscale thermophoresis.

(398) The results showed that compound 2 did not bind with normal ATXN3 protein (MBP-ATXN3-Q28), and the $K_{sub.d}$ of compound 2 with abnormal ATXN3 protein (MBP-ATXN3-Q68) was 2.77 μ M. The compound of the present invention selectively binded to ATXN3 with abnormally expanded polyQ.

Example 10 Effects of the Compounds on ATXN3 Level in Fibroblasts of Patients with Spinocerebellar Ataxia Type 3

(399) SCA3 patient fibroblasts (Q74), wild-type cells (Q27) were treated with 100 nM compound 1 or 50 nM compound 2 for 2 days. Then the levels of mutant ATXN3 protein (ATXN3-Q74, SEQ ID NO: 3) and wild-type ATXN3 protein (ATXN3-Q27, SEQ ID NO: 4) were determined by Western-blots (FIG. 22). A decrease in mutant ATXN3 protein level was observed on SCA3 patient fibroblasts (Q74), but no decrease in wild-type ATXN3 protein level was observed. Therefore, the compound can be used to treat SCA3.

Example 11 Effects of the Compounds on the Level of ATXN1 with Abnormally Expanded polyQ in Cells

(400) (1) PolyQ-ATXN1 cDNA (where polyQ is Q92) was synthesized de novo and subcloned into pcDNA vector. It transfected into HEK293T cells (ATCC, CRL-3216) to obtain cells expressing His-ATXN1-Q92.

(401) (2) Cells prepared in (1) were treated with 100 nM compound (compound 1, or compound 2, or compound 4, or compound 6) for 2 days. Then the level of mutant ATXN1 protein (His-ATXN1-Q92, SEQ ID NO: 5 with His tag at the N-terminal) was detected by HTRF (antibody pair: anti-His-Tb/MW1-D2, FIG. 23). Decreased levels of mutant ATXN1 were observed on fibroblasts (Q92) of SCA1 patients. Therefore, the compound of the present invention can be used to treat or prevent spinocerebellar ataxia type 1.

(402) TABLE-US-00004 Sequence Listing Q25-GFP: [SEQ ID NO: 1]

MQQQQQQQQQQQQQQQQQQQQQQQQQQQQQMSKGEELFTGVVPILVELDGDVNGH
KFSVRGEGEGDATNGKLTCLKFICTTGKLPVPWPTLVTTLTLYGVQCFSRYPDHMKRHD
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LEYNFNSHNVYITADKQKNGIKANFKIRHNVEDGSVQLADHYQQNTPIGDGPVLLPD
NHYLSTQSVLSKDPNEKRDHMLLEFVTAAGITHGGSG HTT-Q23: [SEQ ID NO: 2]
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EIKKNGAPRSLRAALWRFAELAHVLRPQKCRPYLVNLLPCLTRTSKRPEESVQETLAA
AVPKIMASFGNFANDNEIKVLLKAFIANLKSSSPTIRRTAAGSAVSICQHSRRTQYFYS
WLLNVLLGLLVPVEDEHSTLLILGVLLTLRYLVPLLQQQVKDTSLKGSFGVTRKEMEY
SPSAEQLVQVYELTLHHTQHHDHNVVTGALELLQQLFRTPPELLQTLTAVGGIGQLT
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DSSVDKFVLRDEATEPGDQENKPCRIKGDIGQSTDDDSAPLVHCVRLLSASFLLTGK
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IDHGDQPVRGATAILCGTLICSILSRSRFHVGDWMGTIRTTLTGNTFSLADCIPLLRKTLK
DESSVTCKLACTAVRNCVMSLCSSSYSELGLQLIIDVLTLRNSSYWLVRTELLETAEI
DFRLVSFLEAKAENLHRGAHHYTGLLKLQERVLNNVVIHLLGDEDPRVRHVAAASLI
RLVPKLFYKCDQGQADPVVAVARDQSSVYLKLLMHETQPPSHFSVSTITRIYRGYNLL
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WASEEEANPAATKQEEVWPALGDRALVPMVEQLFSLHLLKVINICAHVLDDEVAPGPAIK
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KIIQLCDGIMASGRKAVTHAIPALQPIVHDLFVLRGTNKADAGKELETQKEVVVSMML
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LNTLFEILAPSSLRPVDMLLRSMFVTPNTMASVSTVQLWISGILAILRVLISQSTEDIVL
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GLAQRHQRLYSLLDRLSTMQDSLSPSPVSSHPLDGDGHVSLETVSPDKDWYVHL
VKSQCWTRSDSALLEGAEVLNRIPAEDMNAFMMNSEFNL SLLAPCLSLGMSEISGGQ
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PKAISEEEEEVDPNTQNPKYITAACEMVAEMVESLQSVLALGHKRNSGVPAFLTPLL
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REEEWDEEEEEEEADAPAPSSPPTSPVNSRKHRAAGVDIHSCSQFLELYSRWILPSSAR
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YLLSNLKGIAHCVNIHSQQHVLVMCATAFYLIENYPLDVGPEFSASIIQMCGVMLSGS
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CMYTGKEKVSPGRTSDPNPAAPDSESVIVAMERVSFLFDRIRKGFPCEARVVARILPQF
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SNFTQRAPVAMATWSLSCFFVSASTSPWVAAILPHVISRMGKLEQVDVNLFCLVATDF
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ID NO: 3]

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DQLLQMIRVQQMHRPKLIGEELAQLKEQRVHKTDLERMLEANDGSGMLDEDEEDLQ
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DQLLQMIRVQQMHRPKLIGEELAQLKEQRVHKTDLERMLEANDGSGMLDEDEEDLQ
RALALSRQEIDMEDEEADLRRAIQLSMQGSSRNISQDMTQTSGTNLTSEELRKRREAY
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VSPVQYAHLPHTFQFIGSSQYSGTYASFIPSQLIPTANPVTSAVASAAGATTPSQRSQLE
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QQ
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AIQAKEVLNGEMEKSRRYGAPSSADLGLGKAGGKSVHPYESSRHVVVHPSPSDYSSR
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KIDSSTVERIEDSHSPGVAVIQFAVGEHRAQVSVEVLVEYPPFFVFGQGWSGCCPERTSQ
LFDLPCSKLSVGDCISLTLKNLKNGSVKKGQPVPDPASVLLKHASKADGLAGSRHRYA
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EKSEDEPPLTLPKPSLIPQEVKICIEGRSNVVK

Claims

1. A method for treating a polyQ-related neurodegenerative disorder in a subject, the method comprising administering an effective amount of a compound or a pharmaceutically acceptable salt thereof to the subject, wherein the polyQ-related neurodegenerative disorder is Huntington's disease; and the compound is selected from: ##STR00083## ##STR00084##
 2. The method of claim 1, wherein the compound or the pharmaceutically acceptable salt thereof is administered in a pharmaceutical composition, and the pharmaceutical composition comprises the compound or the pharmaceutically acceptable salt and at least one pharmaceutically acceptable carrier.
 3. A method for treating a polyQ-related neurodegenerative disorder in a subject, the method comprising administering an effective amount of a compound, or a pharmaceutically acceptable salt thereof, to the subject, wherein the polyQ-related neurodegenerative disorder is spinocerebellar ataxia type 1; and the compound is selected from: ##STR00085##
 4. The method of claim 3, wherein the compound or the pharmaceutically acceptable salt thereof is administered in a pharmaceutical composition, and the pharmaceutical composition comprises the compound or the pharmaceutically acceptable salt and at least one pharmaceutically acceptable carrier.
 5. A method for treating a polyQ-related neurodegenerative disorder in a subject, the method comprising administering an effective amount of a compound, or a pharmaceutically acceptable salt thereof, to the subject, wherein the polyQ-related neurodegenerative disorder is spinocerebellar ataxia type 3; and the compound is: ##STR00086##
 6. The method of claim 5, wherein the compound or the pharmaceutically acceptable salt thereof is administered in a pharmaceutical composition, and the pharmaceutical composition comprises the compound or the pharmaceutically acceptable salt and at least one pharmaceutically acceptable carrier.
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